SMT Z.S Patel College of Computer Application
Affiliated to
Veer Narmad South Gujarat University, Surat.

Project Report on,

QuizMaster

Challenge Your Mind

As a Partial Requirement for Degree of

Bachelor of Computer Application

(BCA) 2025 – 2026

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Project Guide	Dhvani Ma'am	

Acknowledgement

I would like to express my sincere gratitude to everyone who supported me throughout the completion of this project.

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I would also like to extend my thanks to **SMT Z.S Patel College of Computer Application** for providing the necessary resources, facilities, and environment to carry out this work.

My heartfelt appreciation goes to my faculty members, friends, and classmates for their encouragement, suggestions, and assistance whenever needed.

Finally, I would like to thank my family for their constant motivation, patience, and moral support throughout this journey.

QUIZMASTER

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Chapter 1: Introduction

1.1 Project Profile

1.2 Overview of Project

1.1 Project Profile

Project Title	QuizMaster
Domain	Challenge Your Mind
Technology Stack	PHP/MySQL Stack (LAMP/WAMP/XAMPP)
Fronted	HTML5, CSS3, Vanilla JavaScript, Bootstrap 5 (v5.3.0)
Backend	Native PHP (v8.x recommended)
Database	MySQL (via PDO)
Tools	VS Code, Git, XAMPP

1.2 Overview of Project

QuizMaster serves as a comprehensive, dynamic web-based assessment platform specifically tailored for educational institutions, corporate training environments, and independent learners. In an era where continuous learning is paramount, QuizMaster bridges the gap between traditional test-taking and modern, interactive digital experiences. The platform is designed to offer a seamless quiz-taking experience that is both engaging for the student and deeply analytical for the administrator or educator. By leveraging established, reliable web technologies, it ensures that users enjoy fast load times, accessible interfaces, and a reliable connection regardless of their device.

The core philosophy behind QuizMaster is "Intuitive Assessment." This design principle influences every aspect of the user interface, from the clean typography and glassmorphism-inspired color palettes to the straightforward navigation and immediate feedback mechanisms. Unlike clunky, legacy Learning Management Systems (LMS) that often feel utilitarian and dated, QuizMaster prioritizes visual harmony and effective use of modern CSS trends. This approach not only elevates the perceived professionalism of the assessments but also reduces

cognitive load for the test-taker, allowing them to focus entirely on demonstrating their knowledge.

From a technical perspective, QuizMaster is built upon the robust and time-tested PHP/MySQL stack, demonstrating a commitment to accessibility, wide server compatibility, and rapid deployment. This architecture provides a straightforward development environment that enhances maintainability. The use of native PHP on the backend, communicating securely with a relational MySQL database via PHP Data Objects (PDO), ensures a stable, synchronous environment capable of handling user sessions, processing complex scoring algorithms, and generating dynamic content on the fly.

Security and data integrity are paramount in the QuizMaster ecosystem. The platform implements industry-standard security measures, including strictly managed PHP sessions and secure, time-limited tokens for password recovery, to ensure that user data is private. Sensitive data, such as user passwords, are hashed using bcrypt before storage, complying with best practices for data protection. Furthermore, the use of parameterized SQL queries via PDO ensures that the database is shielded from SQL injection vulnerabilities, maintaining the reliability of the assessment records.

Finally, QuizMaster is designed with functional expansion in mind. The architecture allows for the straightforward integration of new features, such as the recently added PHPMailer for automated notifications and the FPDF library for dynamic certificate generation. Whether it is adding new question types, integrating advanced reporting tools, or expanding the user base, the underlying system is engineered to handle increased complexity while maintaining its core focus on reliable, accessible online assessment.

Chapter 2:
Proposed System

2.1 Objectives
2.2 Hardware & Software Platforms
2.3 Scope of Projects

2.1 Objectives

The primary objective of the QuizMaster platform is to modernize and streamline the process of knowledge assessment. In an educational landscape increasingly reliant on digital tools, having an intuitive and reliable testing platform is critical. QuizMaster aims to provide a visually engaging interface that encourages participation and reduces test anxiety. By moving away from paper-based or archaic digital testing systems, the platform seeks to create a dynamic environment that resonates with modern learners, thereby improving completion rates and assessment accuracy.

A key operational objective is to simplify the complex processes involved in quiz creation and result management. For administrators and educators, the platform is designed to be a powerful tool that simplifies daily tasks such as drafting multiple-choice questions, organizing content into specific categories, and grading attempts. By automating the scoring and organizing these workflows, QuizMaster aims to reduce manual grading errors, save educators immense amounts of time, and allow them to operate more efficiently.

Enhancing user engagement and providing immediate value are central to the platform's user-centric goals. QuizMaster strives to create a rewarding user journey, from the moment a student logs in to the instant they complete a quiz. This involves implementing clear progress indicators, immediate automated scoring on submission, and tangible rewards for success. By minimizing friction and offering immediate feedback, including downloadable PDF certificates for passing scores, the system encourages continued learning and improves overall user satisfaction.

Ensuring reliability and data integrity is a fundamental architectural objective. As an assessment tool, its software infrastructure must be able to securely handle concurrent test-takers without losing data or crashing. QuizMaster is engineered to be robust, utilizing the proven stability of Apache/Nginx web servers and relational MySQL databases. The goal is to provide a platform that

institutions can trust to accurately record and maintain student performance histories over time.

Another significant objective is to ensure the security and privacy of student data. With the increasing requirements of educational privacy laws, building trust with users is essential. QuizMaster aims to implement rigorous security protocols, including secure password hashing and token-based account recovery mechanisms. The objective is to safeguard sensitive student information, ensuring a secure environment for examination.

Lastly, the project aims to provide a data-driven foundation for educational decision-making. By capturing detailed information on quiz attempts, question accuracy, and overall scores, the system creates a valuable repository of performance data. The objective is to empower educators with insights that help them understand student comprehension, identify difficult topics, and tailor their teaching strategies for better educational outcomes.

2.2 Hardware & Software Platforms

Hardware Development Environment:

Minimum:	• **Processor:** Single-core Processor (1 GHz+) • **RAM:** 1GB (2GB recommended for local servers like XAMPP) • **Storage:** 100MB for application files, plus Database storage (HDD/SSD)
Recommended:	• **Processor**: Modern Multi-core Processor • **RAM:** 4GB minimum recommended • **Storage:** 512GB SSD for faster database I/O

Software Development Environment:

Operating System:	Windows 10/11, macOS 10.15+, or Linux.
Frontend	HTML5, CSS3, Vanilla JavaScript, Bootstrap 5 (v5.3.0)
Backend	Native PHP (v8.x recommended)
Database:	MySQL (via PDO)
Code Editor:	Visual Studio Code (Recommended).
Browser:	Chrome, Firefox, or Edge.
Version Control:	Git.

2.3 Scope of Project

The scope of QuizMaster encompasses a comprehensive Assessment Discovery module designed to facilitate an effortless learning experience. This includes a dynamic, paginated quiz catalog that allows students to browse through available tests with descriptions and metadata. The system supports category filtering and keyword search capabilities, enabling students to quickly find quizzes relevant to their current studies or interests, ensuring that the question bank is easily accessible and navigable.

User Management is another critical component within current scope. The system provides a secure and user-friendly registration and login process, featuring backend email validation and strong password hashing. Once authenticated, users have access to a personalized dashboard where they can manage their profiles, update security settings, and view their accumulated experience points (XP). The scope also includes a robust, secure "Forgot Password" flow utilizing 15-minute expiring tokens sent via email (PHPMailer integration).

Assessment Processing forms the core of the educational functionality. The project scope covers the entire lifecycle of a quiz attempt, starting from an interactive quiz interface that tracks time limits. This workflow collects user answers, processes them immediately upon submission against the database schema, and calculates a final score. Furthermore, the system tracks the complete attempt history for each user, providing them with a visual timeline of their past performances, pass/fail statuses, and the ability to review their detailed correct/incorrect answers.

On the administrative side, the scope includes a powerful Admin Dashboard. This restricted area provides educators with full control over the platform's content and operations. Admins can perform CRUD (Create, Read, Update, Delete) operations on Categories, Quizzes, and individual Questions (Multiple Choice, True/False). The dashboard also facilitates Student Management, allowing admins

to view registered users, monitor their activities, and manage platform-wide settings.

The security scope involves implementing fundamental measures to protect the platform and its users. This includes the implementation of strict PHP session management for stateful and secure access control, ensuring students cannot access admin routes. Input validation and sanitation on both the client (JS) and server (PHP) sides are included to prevent common web vulnerabilities. Specifically, all database interactions enforce Prepared Statements (PDO) to eradicate SQL injection risks.

Finally, the technical scope defines the boundaries of the platform's operation. QuizMaster is designed as a web-based responsive application accessible via standard web browsers on desktop, tablet, and mobile devices. It requires a standard web server equipped with PHP and MySQL. The current scope includes advanced reward mechanisms like on-the-fly PDF Certificate generation (via FPDF) for passing scores, and automated SMTP email notifications. It excludes, for this version, native mobile app development or live proctoring features.

Chapter 3:
System Design

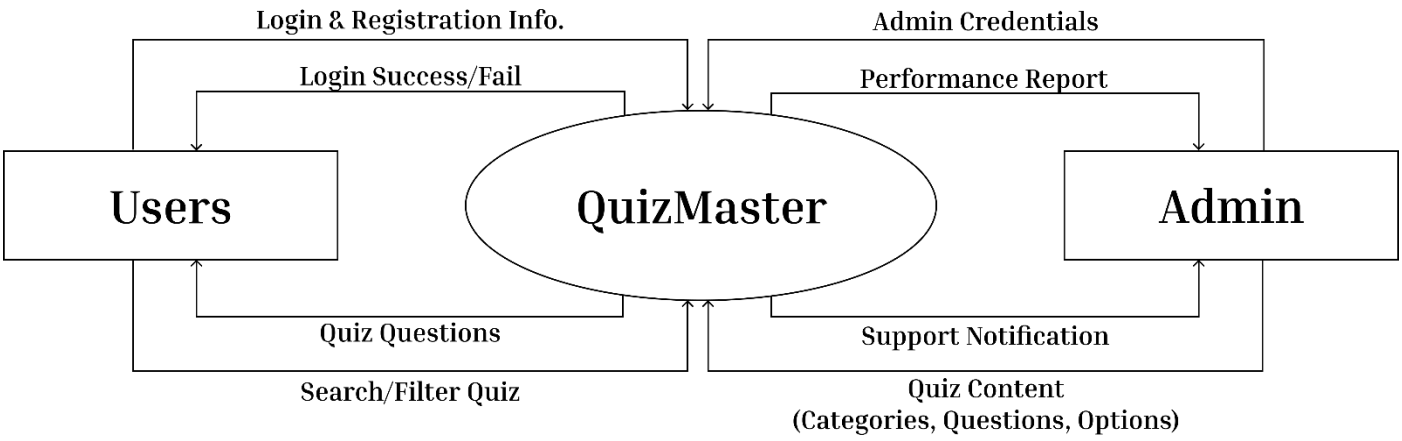
3.1 DFD (Data Flow Diagram)
3.2 UML Diagram
3.3 Data Dictionary
3.4 Interface Design (Screenshots)

3.1 DFD (Data Flow Diagram)

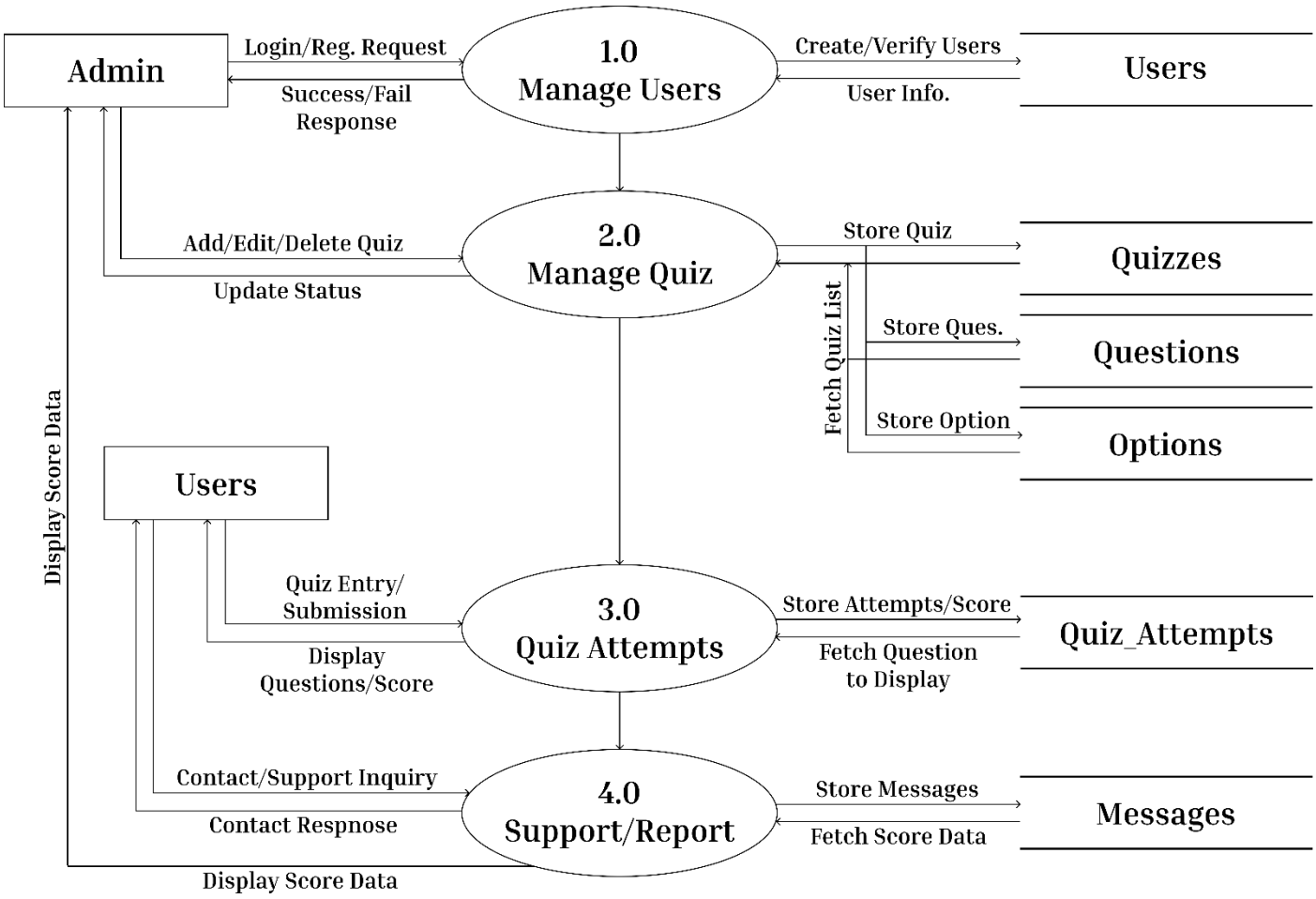
DFD Diagrams:

The Data Flow Diagram (DFD) serves as a visual representation of how information moves through the QuizMaster system. It maps out the flow of data from external entities, such as the Student and the Administrator, into the system's various processes and data stores.

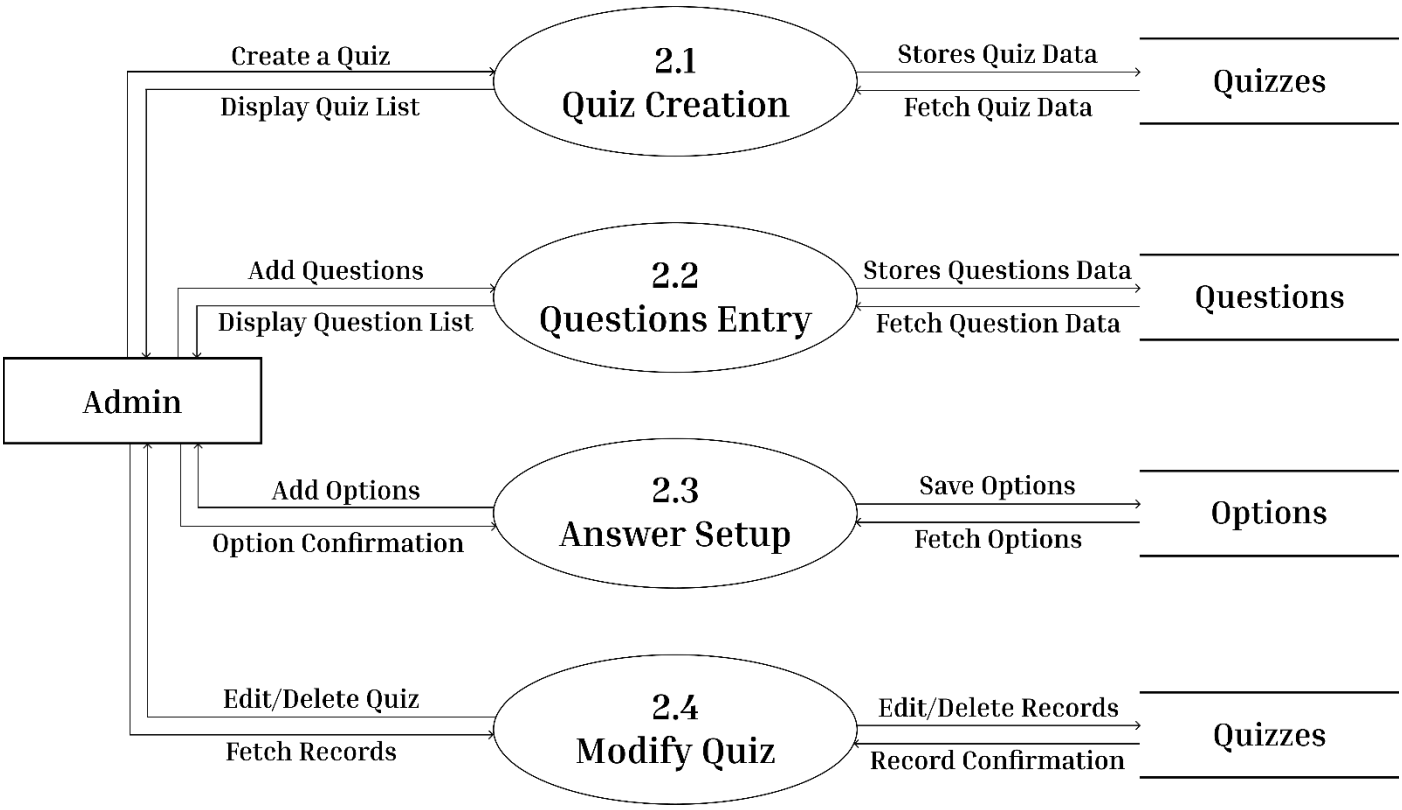
DFD Level 0:



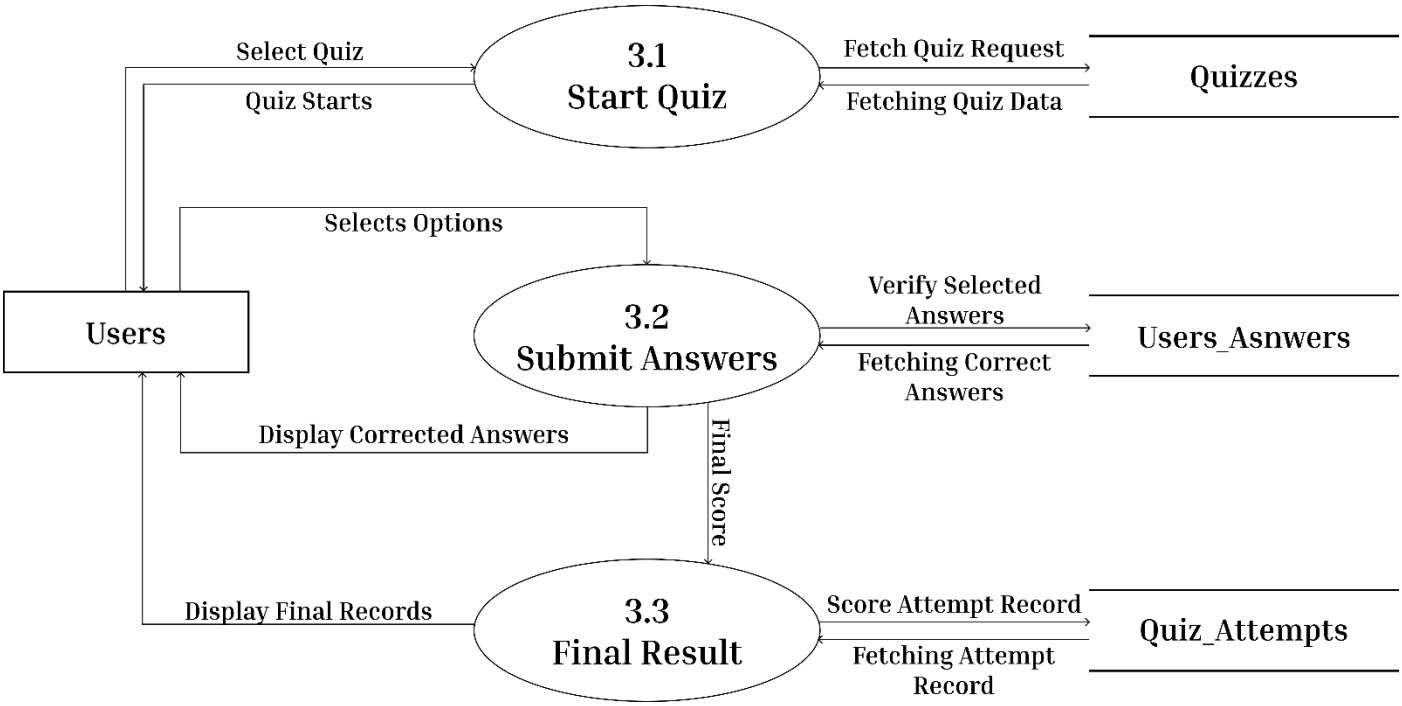
DFD Level 1:


Admin
Login/Reg. Request
Success/Fail Response
1.0 Manage Users
Create/Verify Users
User Info.
Users
Add/Edit/Delete Quiz
Update Status
2.0 Manage Quiz
Store Quiz
Quizzes
Fetch Quiz List
Store Ques.
Questions
Store Option
Options
Display Score Data
Users
Quiz Entry/ Submission
Display Questions/Score
3.0 Quiz Attempts
Store Attempts/Score
Fetch Question to Display
Quiz_Attempts
Contact/Support Inquiry
Contact Respnose
4.0 Support/Report
Store Messages
Fetch Score Data
Messages
Display Score Data

DFD Level 2.1: (Manage Quiz)



DFD Level 3.1: (Quiz Attempts)

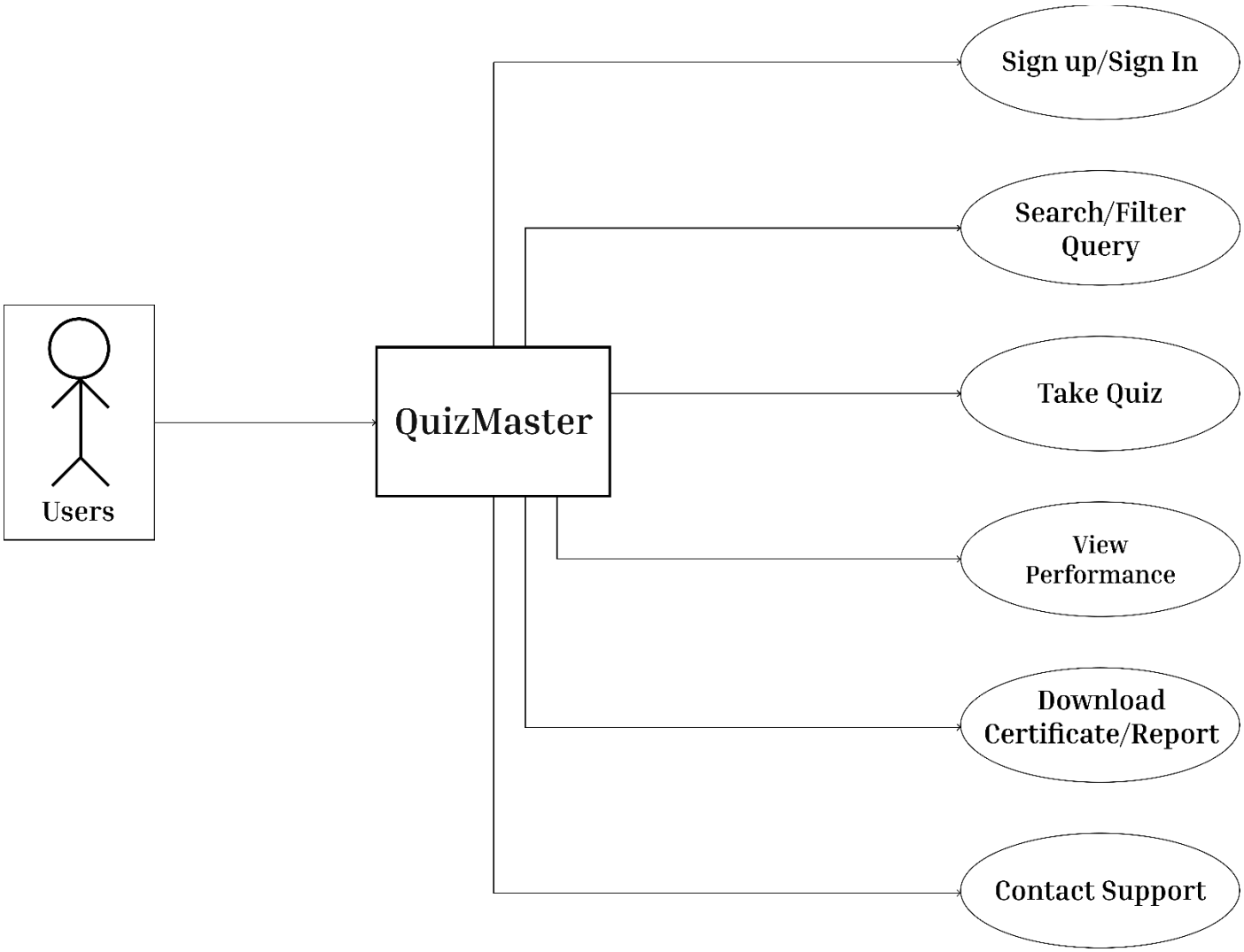


3.2 UML Diagram:

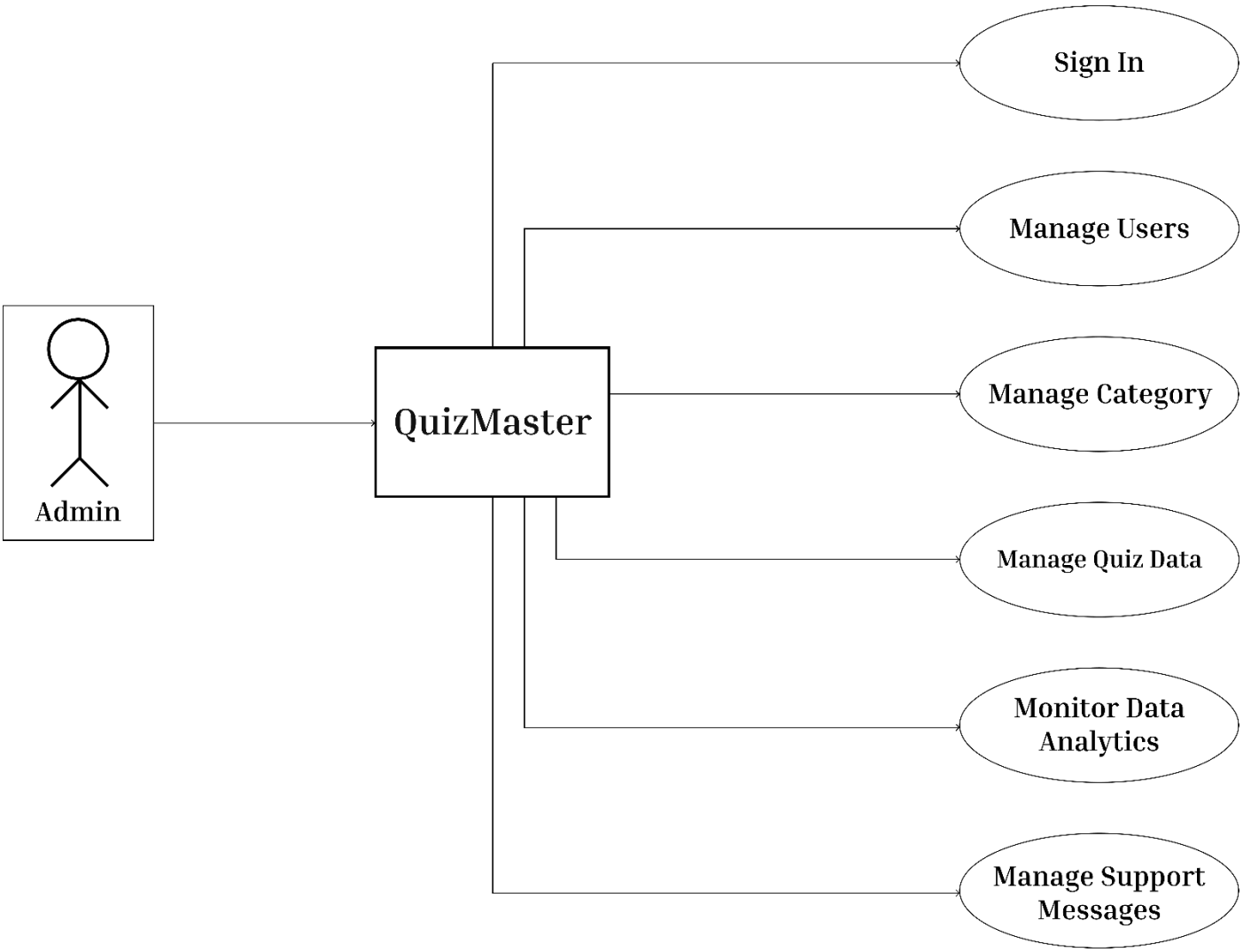
Use Case Diagrams:

In the context of our native PHP API, the UML diagram best translates to an Entity-Relationship (ER) model representing the relational database structure. This blueprint defines the tables, their attributes, and the strict foreign-key relationships between them, ensuring data integrity.

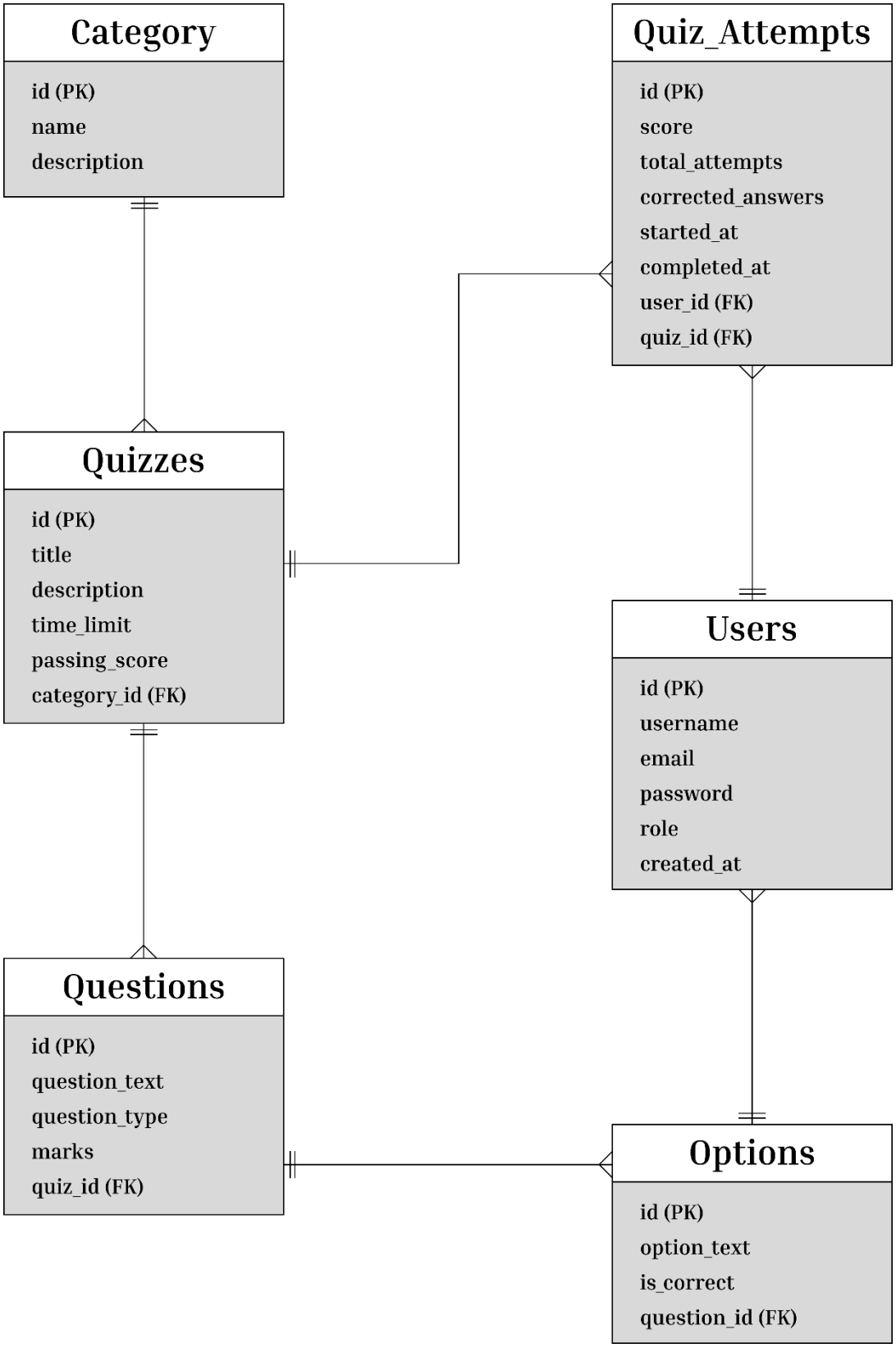
User's Use Case Diagram:



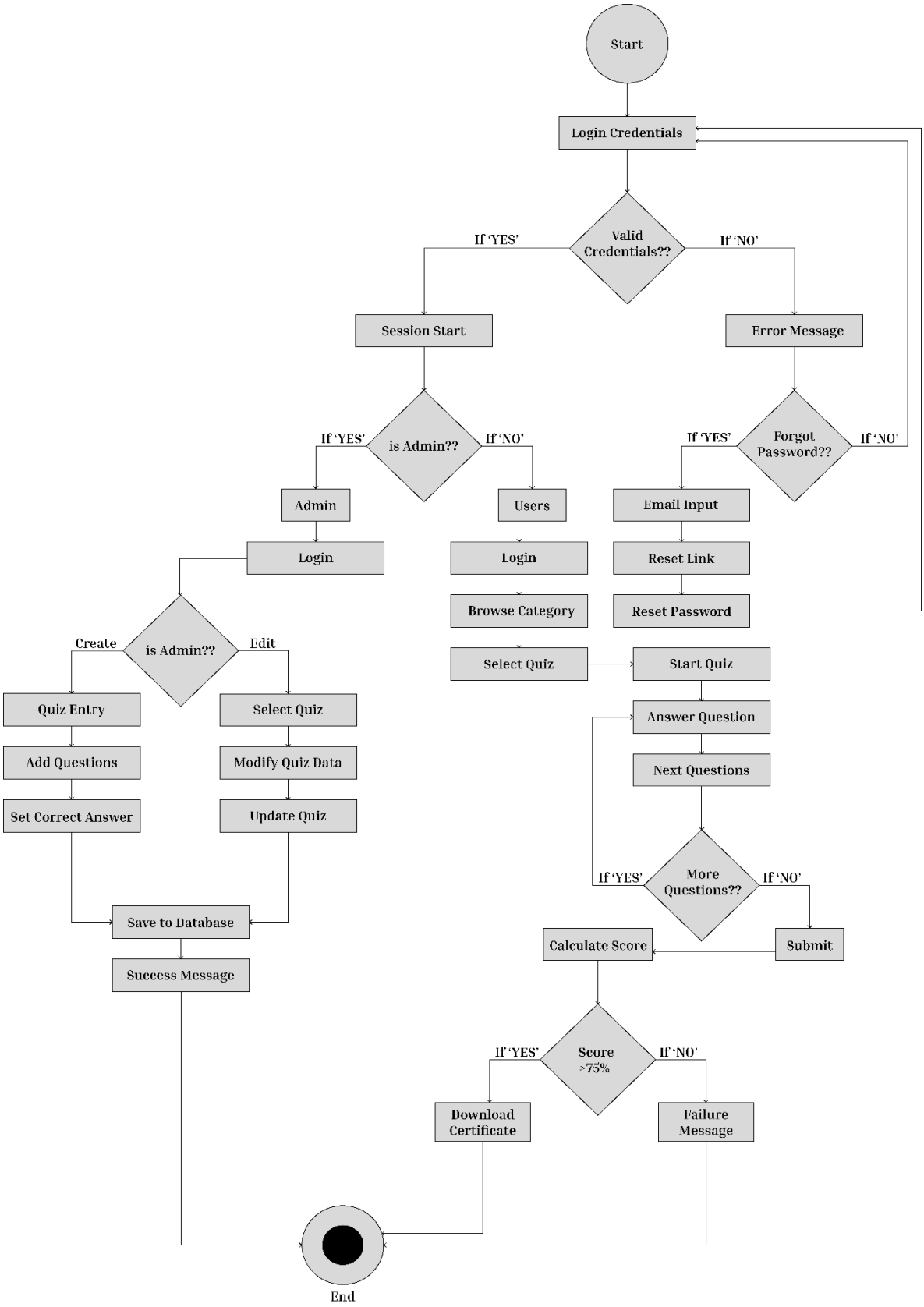
Use Case Diagram: (Admin)



3.2.1 ER Diagram



3.2.2 Activity Diagram



3.3 Data Dictionary

The Data Dictionary serves as a centralized repository of information about data, encompassing its meaning, relationships to other data, origin, usage, and format. In the context of the QuizMaster platform, it provides a structured and comprehensive breakdown of the core database collections implemented in MySQL. This documented vocabulary ensures that all developers and database administrators have a unified understanding of the data architecture, preventing inconsistencies in data definition and usage across different modules of the application.

1). Users Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each user.
username	VARCHAR(50)	Unique, Not Null	User-selected username used for login and identification.
email	VARCHAR(100)	Unique, Not Null	User's primary email address used for communication and password recovery.
password	VARCHAR(255)	Not Null	Hashed password stored using PHP password_hash() (bcrypt).
role	ENUM('admin','student')	Default: 'student'	Defines access level. Admin has full dashboard access; Student can take quizzes.
reset_token	VARCHAR(64)	Nullable	Secure random token used for password reset verification.
reset_token_expires_at	DATETIME	Nullable	Expiration timestamp (15 minutes validity) for password reset token.

2. Quizzes Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each quiz.
title	VARCHAR(255)	Not Null	Name of the quiz displayed to students.
description	TEXT	Nullable	Description of quiz subject or instructions.
category_id	INT	Foreign Key	References categories.id.
time_limit	INT	Default: 0	Time limit in minutes (0 = no time limit).
passing_score	INT	Default: 50	Minimum percentage required to pass the quiz.
created_by	INT	Foreign Key	References users.id (Admin who created the quiz).

3. Questions Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each question.
quiz_id	INT	Foreign Key	References quizzes.id (Cascade on delete).
question_text	TEXT	Not Null	The question displayed to the student.
question_type	ENUM('multiple_choice', 'true_false')	Default: 'multiple_choice'	Determines type of question rendering.
marks	INT	Default: 1	Marks awarded for correct answer.

4. Categories Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for category.
name	VARCHAR(100)	Not Null	Category name (e.g., Mathematics, Science).
description	TEXT	Nullable	Additional details about category.

5. Options Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each answer option.
question_id	INT	Foreign Key	References questions.id.
option_text	TEXT	Not Null	Text of the answer choice.
is_correct	TINYINT(1)	Default: 0	1 = Correct Answer, 0 = Incorrect Answer.

6. Quiz_Attempts Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each quiz attempt.
user_id	INT	Foreign Key	References users.id.
quiz_id	INT	Foreign Key	References quizzes.id.
score	INT	Default: 0	Final calculated score.
completed_at	TIMESTAMP	Nullable	Timestamp when quiz attempt was completed.

7. Users_Answers Table:

Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each answer submission.
attempt_id	INT	Foreign Key	References quiz_attempts.id.
question_id	INT	Foreign Key	References questions.id.
selected_option_id	INT	Foreign Key	References options.id.
is_correct	TINYINT(1)	Default: 0	Snapshot of correctness at submission time.

8. Settings Table:

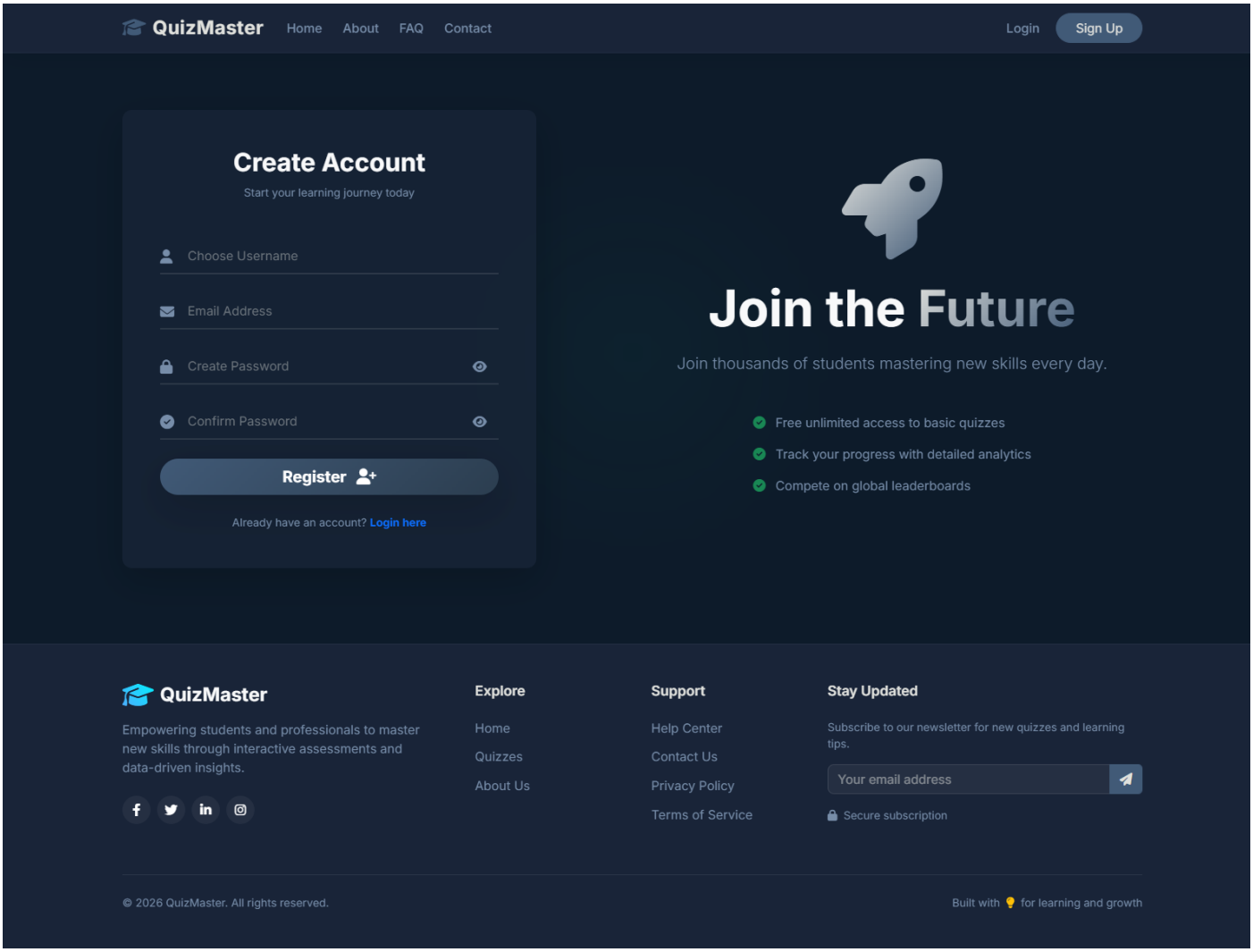
Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for configuration record.
setting_key	VARCHAR(50)	Unique, Not Null	Configuration key (e.g., site_name, maintenance_mode).
setting_value	TEXT	Nullable	Value assigned to the configuration key.
updated_at	TIMESTAMP	Auto Update	Automatically updates when record is modified.

9. Contact_Messages Table:

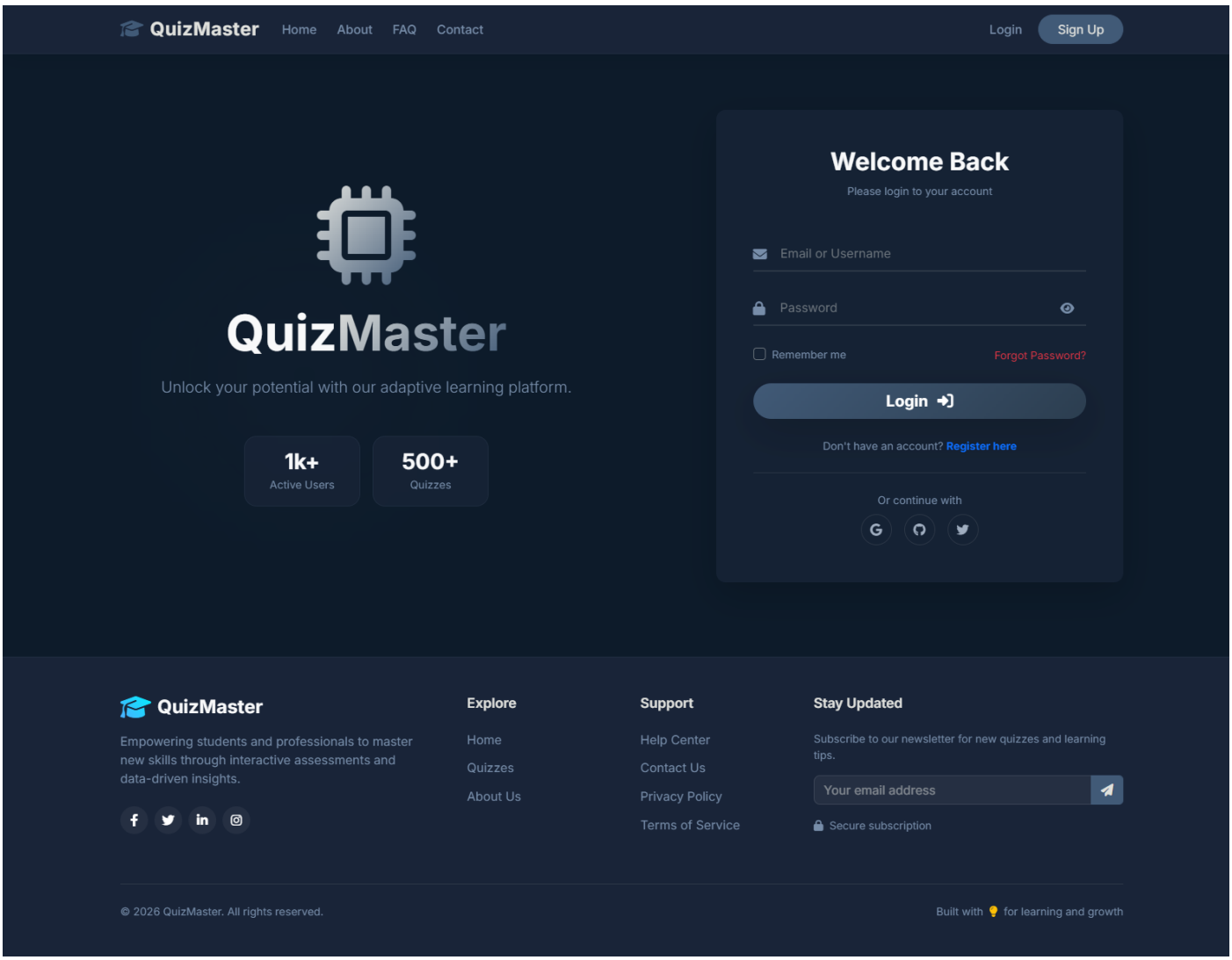
Field	Type	Constraints	Description
id	INT	Primary Key, Auto Increment	Unique identifier for each contact message.
name	VARCHAR(100)	Not Null	Name of the person submitting the message.
email	VARCHAR(100)	Not Null	Email address for reply.
subject	VARCHAR(255)	Not Null	Subject/title of the message.
message	TEXT	Not Null	Detailed content of the message.
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Timestamp when message was submitted.

3.4 Interface (Screenshots)

Registration Page:



Login Page: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Unlock your potential with our adaptive learning platform.
1k+
Active Users
500+
Quizzes
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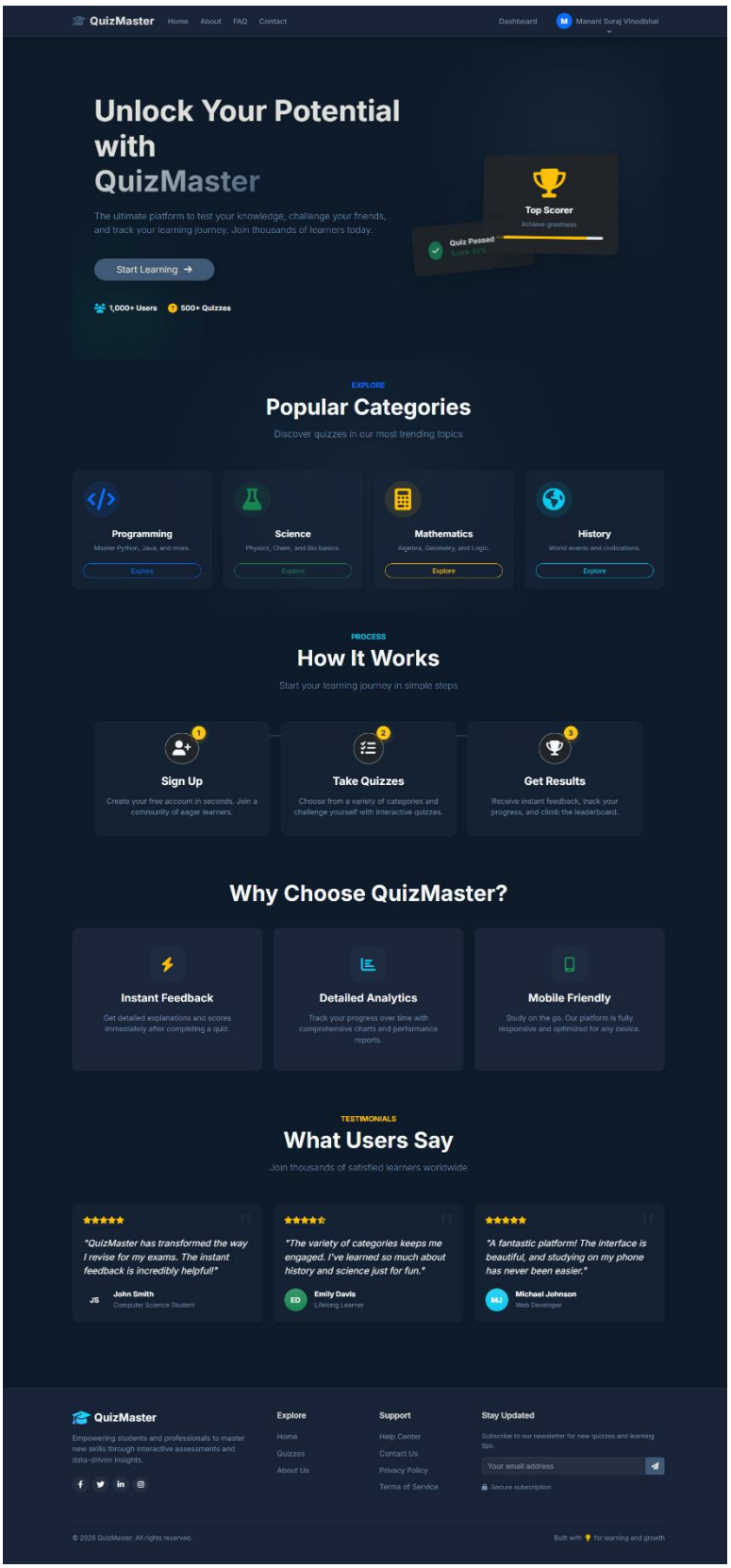
Home Page: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The ultimate platform to test your knowledge, challenge your friends, and track your learning journey. Join thousands of learners today.
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1,000+ Users
500+ Quizzes
Top Scorer
Achieve greatness
Quiz Passed
EXPLORE
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Discover quizzes in our most trending topics
Programming
Master Python, Java, and more.
Explore
Science
Physics, Chem, and lab basics.
Explore
Mathematics
Algebra, Geometry, and Logic.
Explore
History
World events and civilizations.
Explore
PROCESS
How It Works
Start your learning journey in simple steps
Sign Up
Create your free account in seconds. Join a community of eager learners.
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Choose from a variety of categories and challenge yourself with interactive quizzes.
Get Results
Receive instant feedback, track your progress, and climb the leaderboard.
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Get detailed explanations and scores immediately after completing a quiz.
Detailed Analytics
Track your progress over time with comprehensive charts and performance reports.
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Study on the go. Our platform is fully responsive and optimized for any device.
TESTIMONIALS
What Users Say
Join thousands of satisfied learners worldwide
"QuizMaster has transformed the way I revise for my exams. The instant feedback is incredibly helpful!"
JS
John Smith
Computer Science Student
"The variety of categories keeps me engaged. I've learned so much about history and science just for fun."
ED
Emily Davis
Lifelong Learner
"A fantastic platform! The interface is beautiful, and studying on my phone has never been easier."
MJ
Michael Johnson
Web Developer
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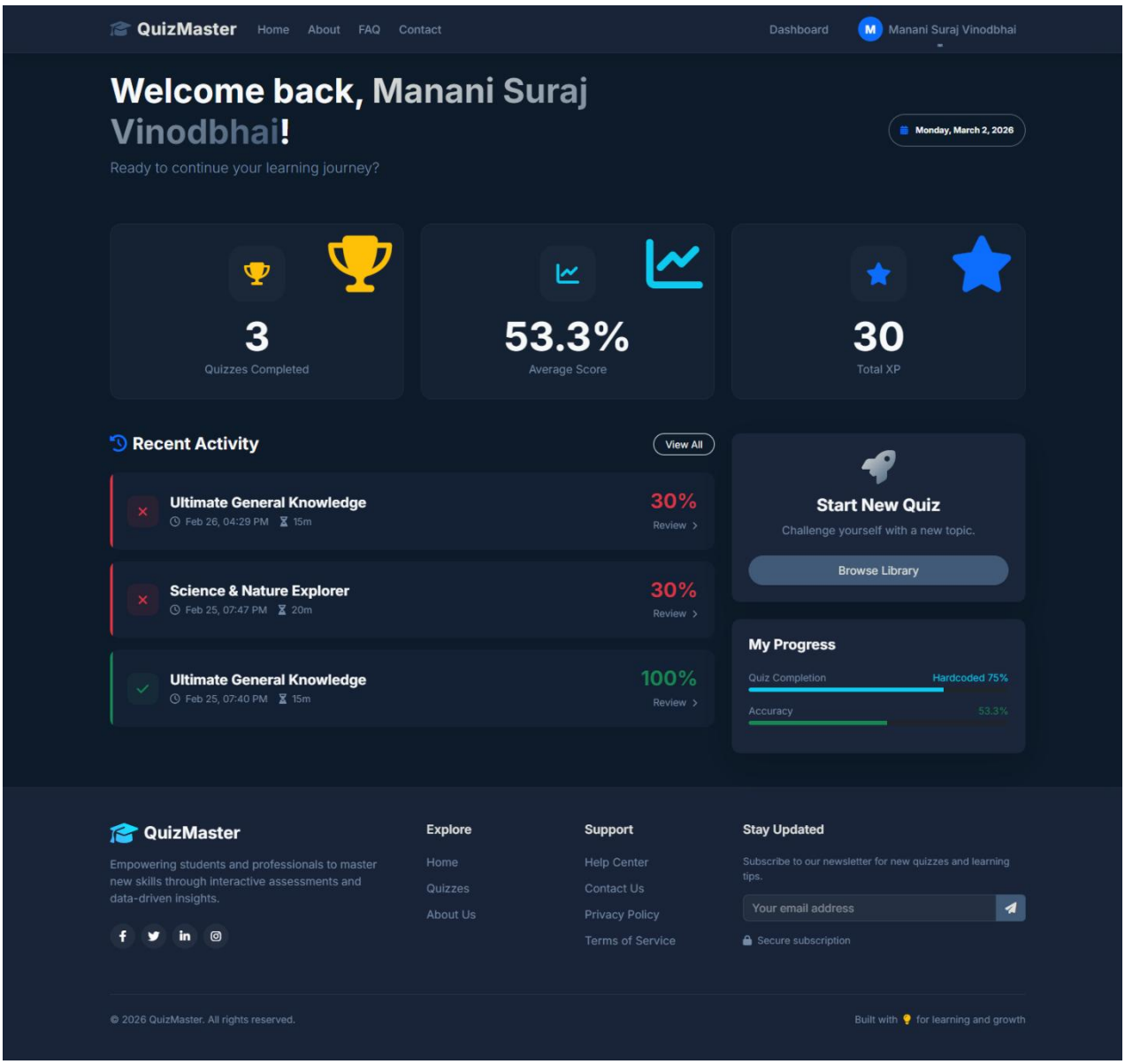
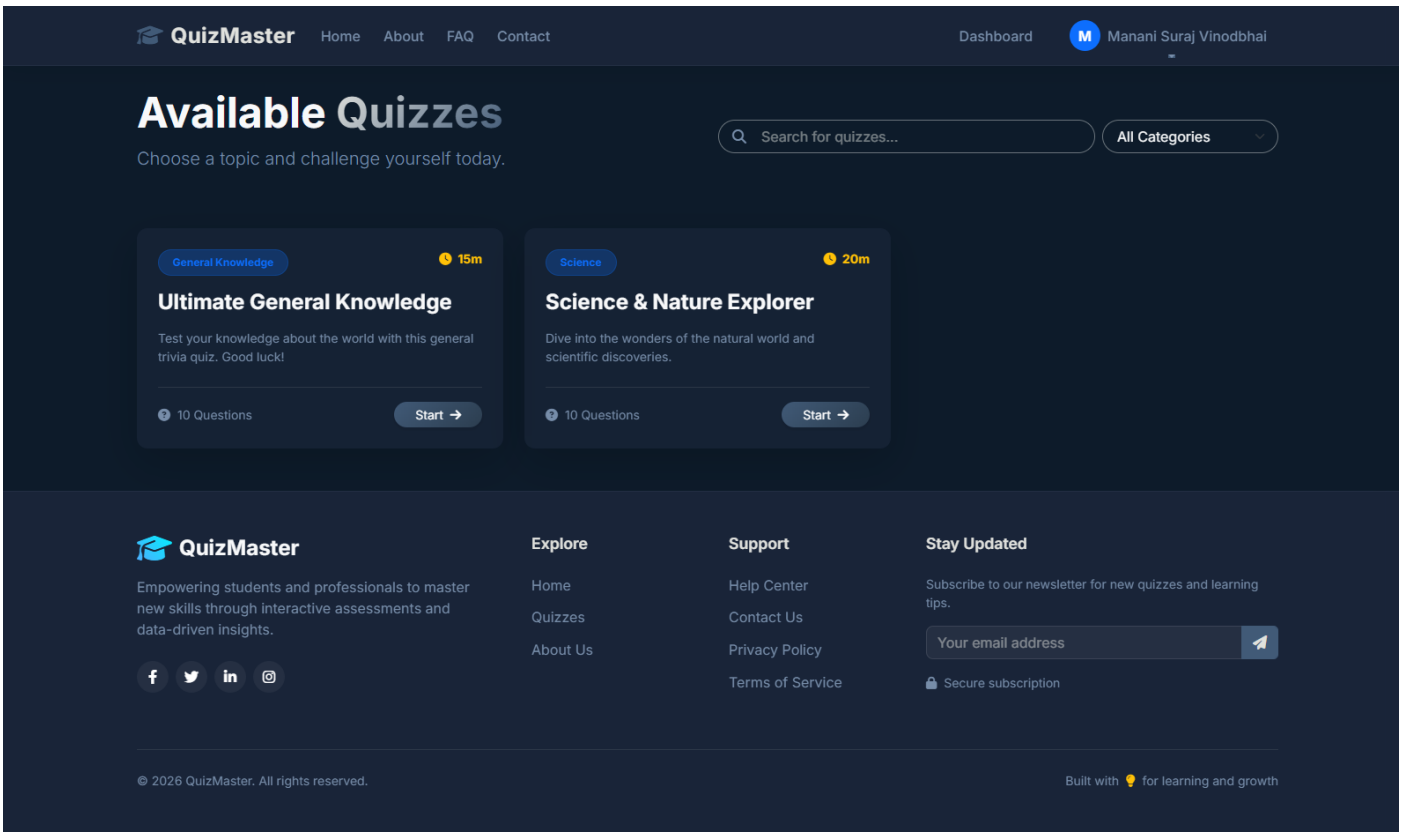
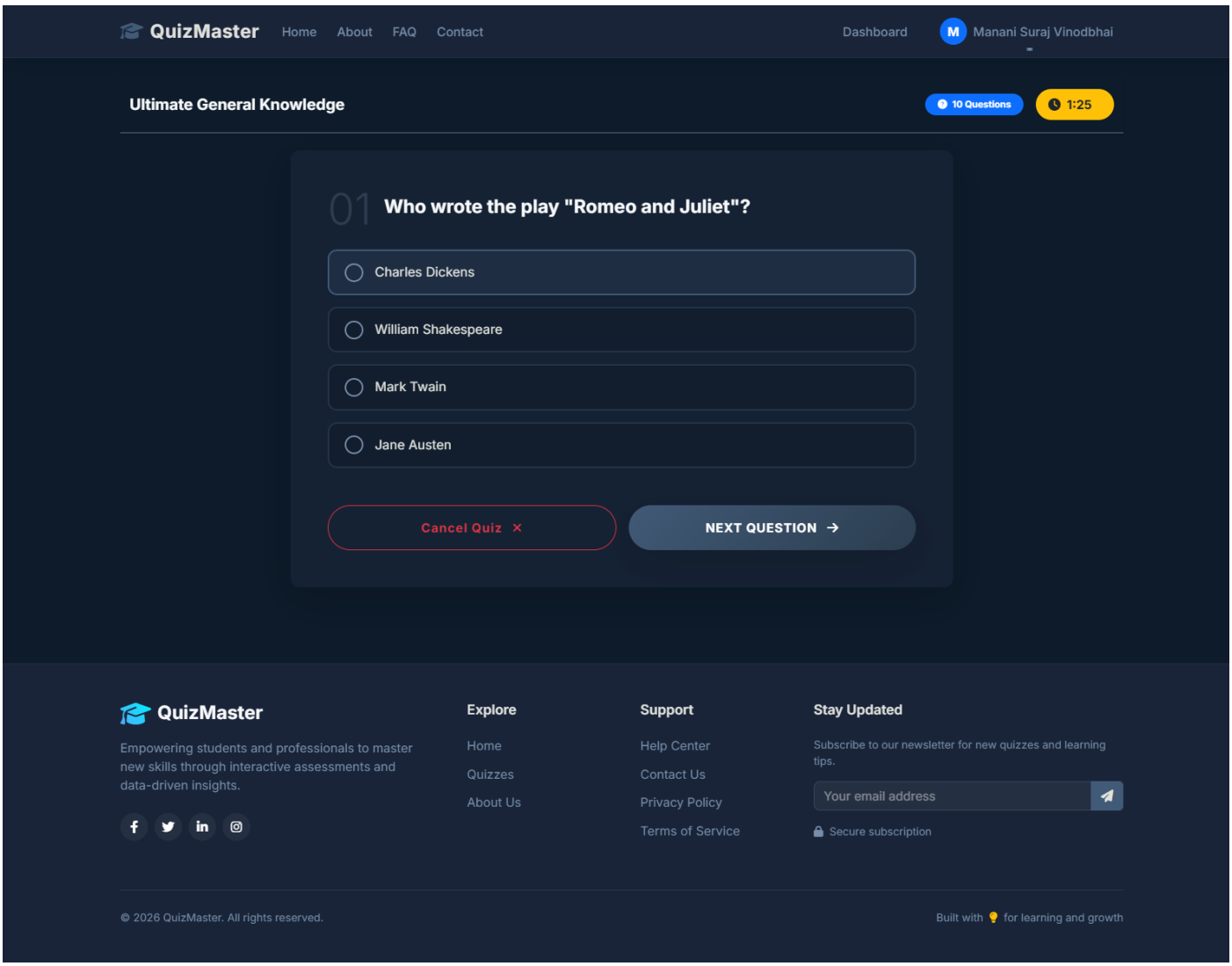
Dashboard Page:



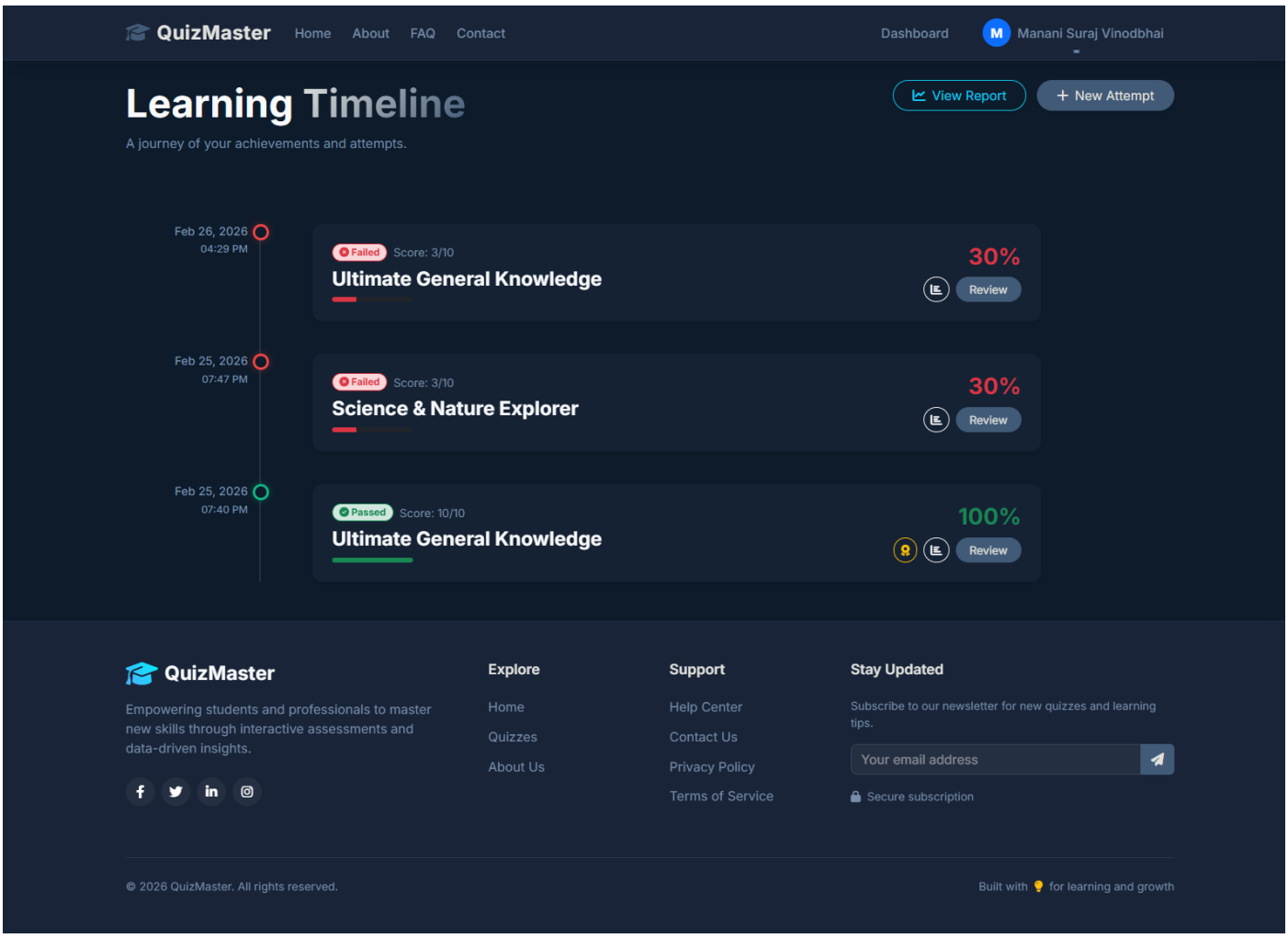
Quizzes Page:



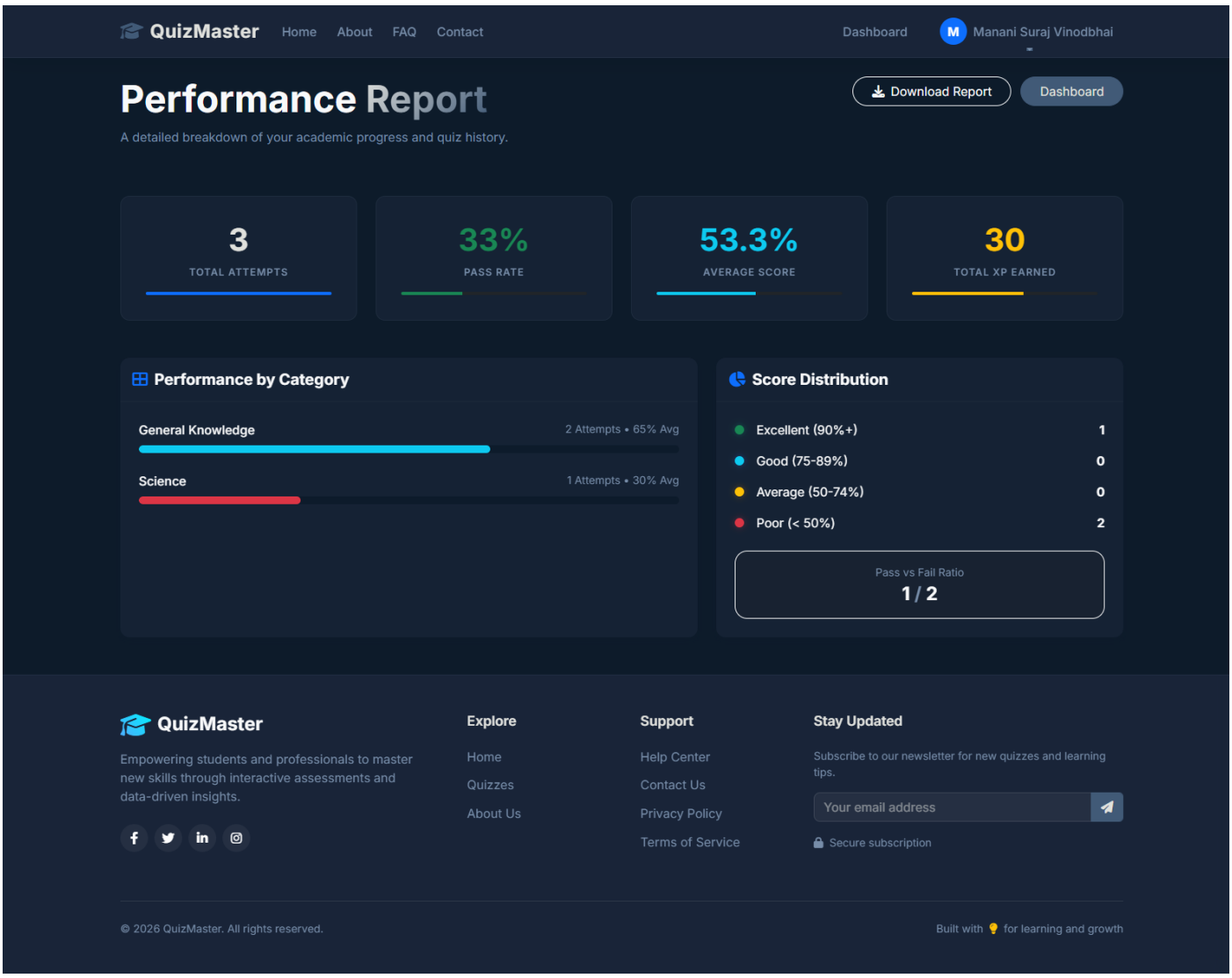
Take-Quiz Page:



History Page:



Report Page: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Performance Report
A detailed breakdown of your academic progress and quiz history.
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Dashboard
3
TOTAL ATTEMPTS
33%
PASS RATE
53.3%
AVERAGE SCORE
30
TOTAL XP EARNED
Performance by Category
General Knowledge
2 Attempts • 65% Avg
Science
1 Attempts • 30% Avg
Score Distribution
Excellent (90%+)
1
Good (75-89%)
0
Average (50-74%)
0
Poor (< 50%)
2
Pass vs Fail Ratio
1 / 2
QuizMaster
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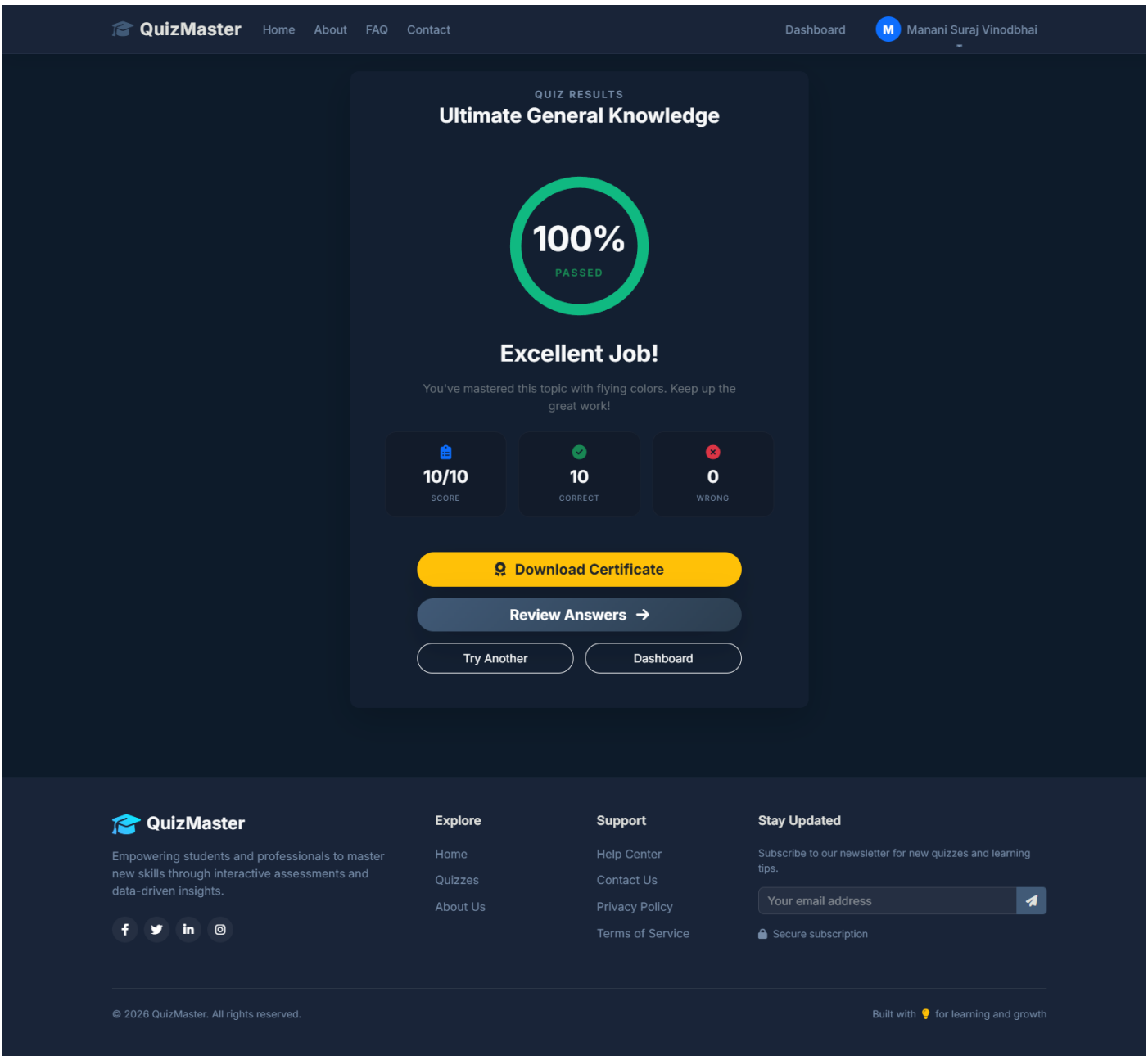
Result Page: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QUIZ RESULTS
Ultimate General Knowledge
100%
PASSED
Excellent Job!
You've mastered this topic with flying colors. Keep up the great work!
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SCORE
10
CORRECT
0
WRONG
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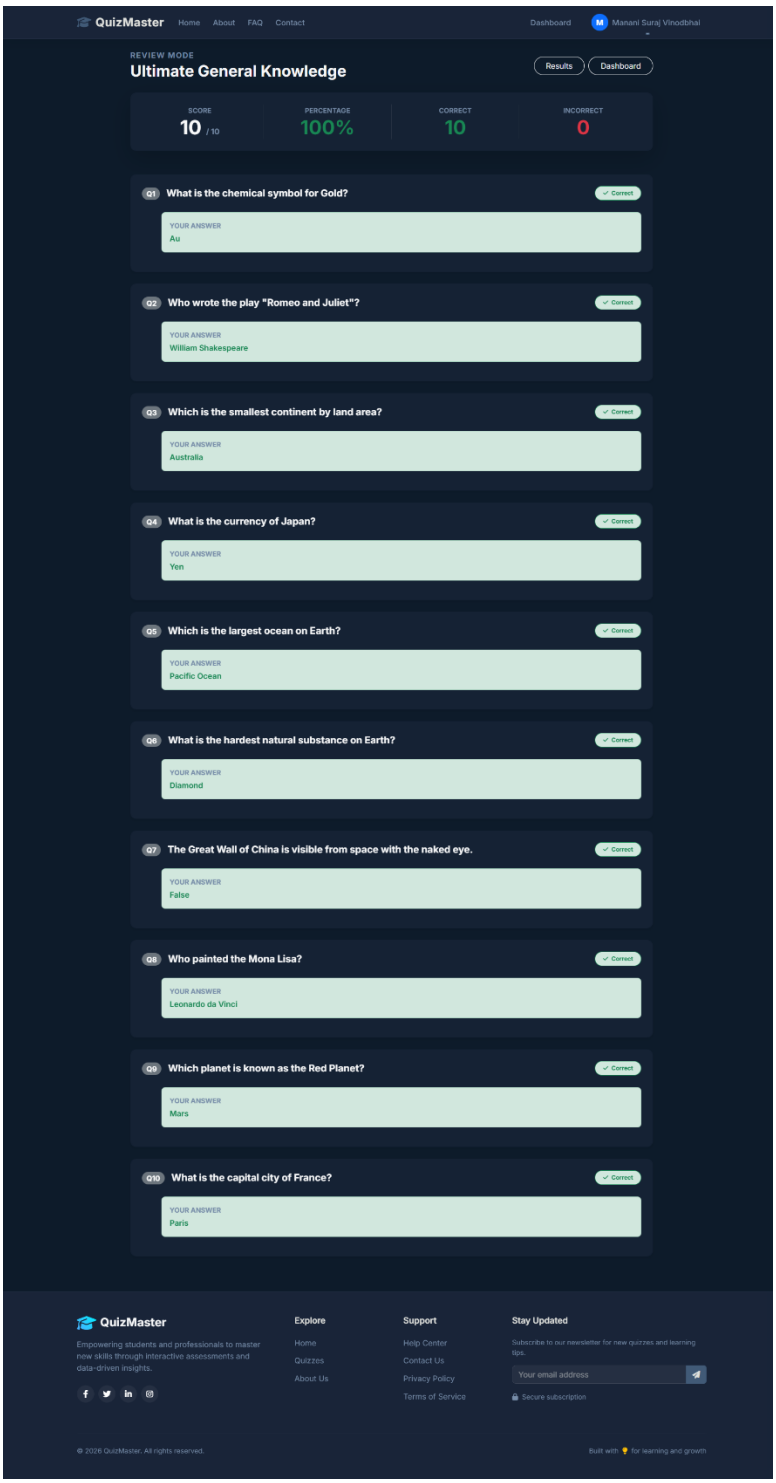
Review Page: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Manani Suraj Vinodbhai
REVIEW MODE
Ultimate General Knowledge
Results
Dashboard
SCORE
10 / 10
PERCENTAGE
100%
CORRECT
10
INCORRECT
0
Q1 What is the chemical symbol for Gold?
Correct
YOUR ANSWER
Au
Q2 Who wrote the play "Romeo and Juliet"?
Correct
YOUR ANSWER
William Shakespeare
Q3 Which is the smallest continent by land area?
Correct
YOUR ANSWER
Australia
Q4 What is the currency of Japan?
Correct
YOUR ANSWER
Yen
Q5 Which is the largest ocean on Earth?
Correct
YOUR ANSWER
Pacific Ocean
Q6 What is the hardest natural substance on Earth?
Correct
YOUR ANSWER
Diamond
Q7 The Great Wall of China is visible from space with the naked eye.
Correct
YOUR ANSWER
False
Q8 Who painted the Mona Lisa?
Correct
YOUR ANSWER
Leonardo da Vinci
Q9 Which planet is known as the Red Planet?
Correct
YOUR ANSWER
Mars
Q10 What is the capital city of France?
Correct
YOUR ANSWER
Paris
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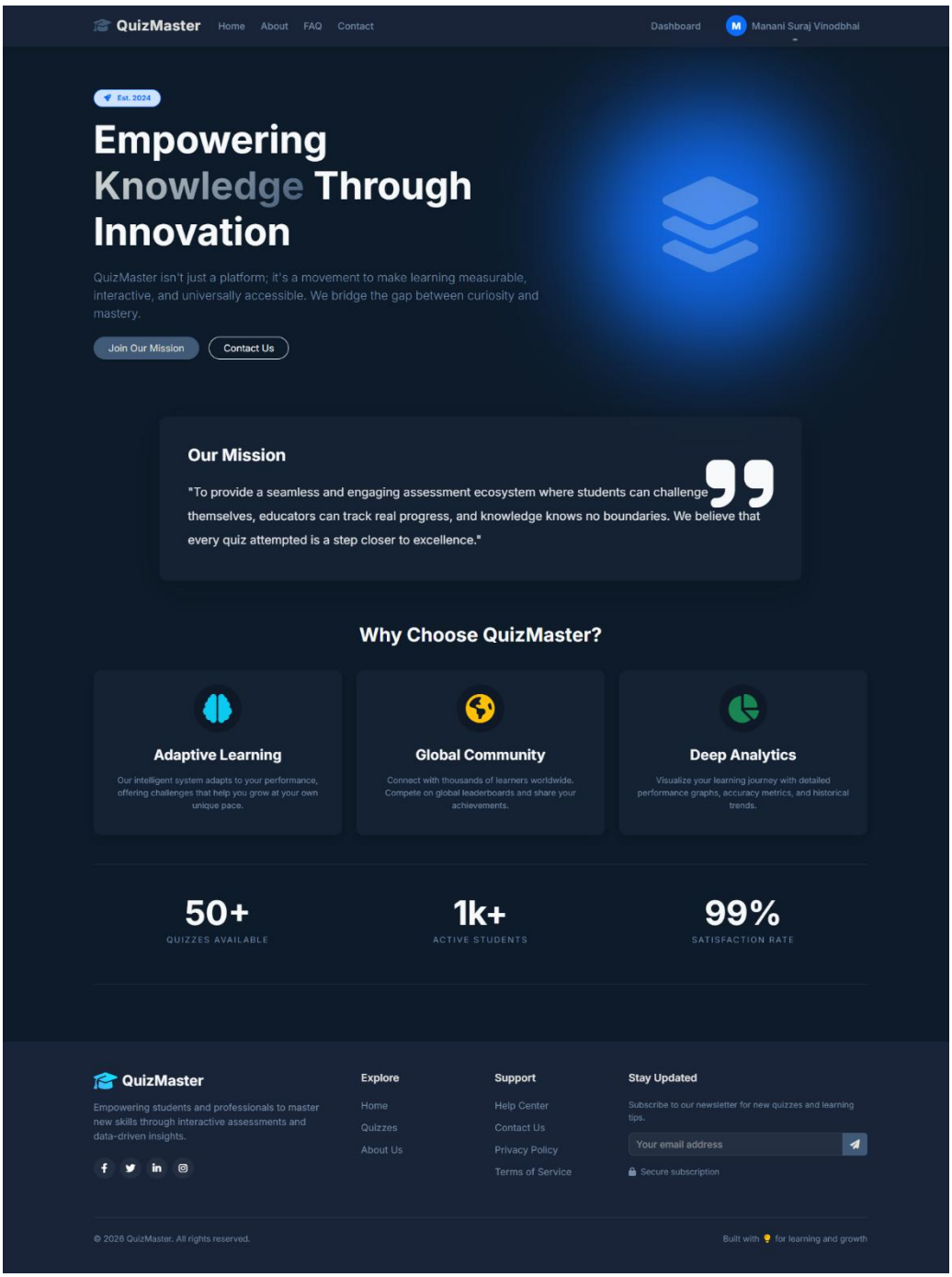
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Est. 2024
Empowering Knowledge Through Innovation
QuizMaster isn't just a platform; it's a movement to make learning measurable, interactive, and universally accessible. We bridge the gap between curiosity and mastery.
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Our Mission
"To provide a seamless and engaging assessment ecosystem where students can challenge themselves, educators can track real progress, and knowledge knows no boundaries. We believe that every quiz attempted is a step closer to excellence."
Why Choose QuizMaster?
Adaptive Learning
Our intelligent system adapts to your performance, offering challenges that help you grow at your own unique pace.
Global Community
Connect with thousands of learners worldwide. Compete on global leaderboards and share your achievements.
Deep Analytics
Visualize your learning journey with detailed performance graphs, accuracy metrics, and historical trends.
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1k+
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SATISFACTION RATE
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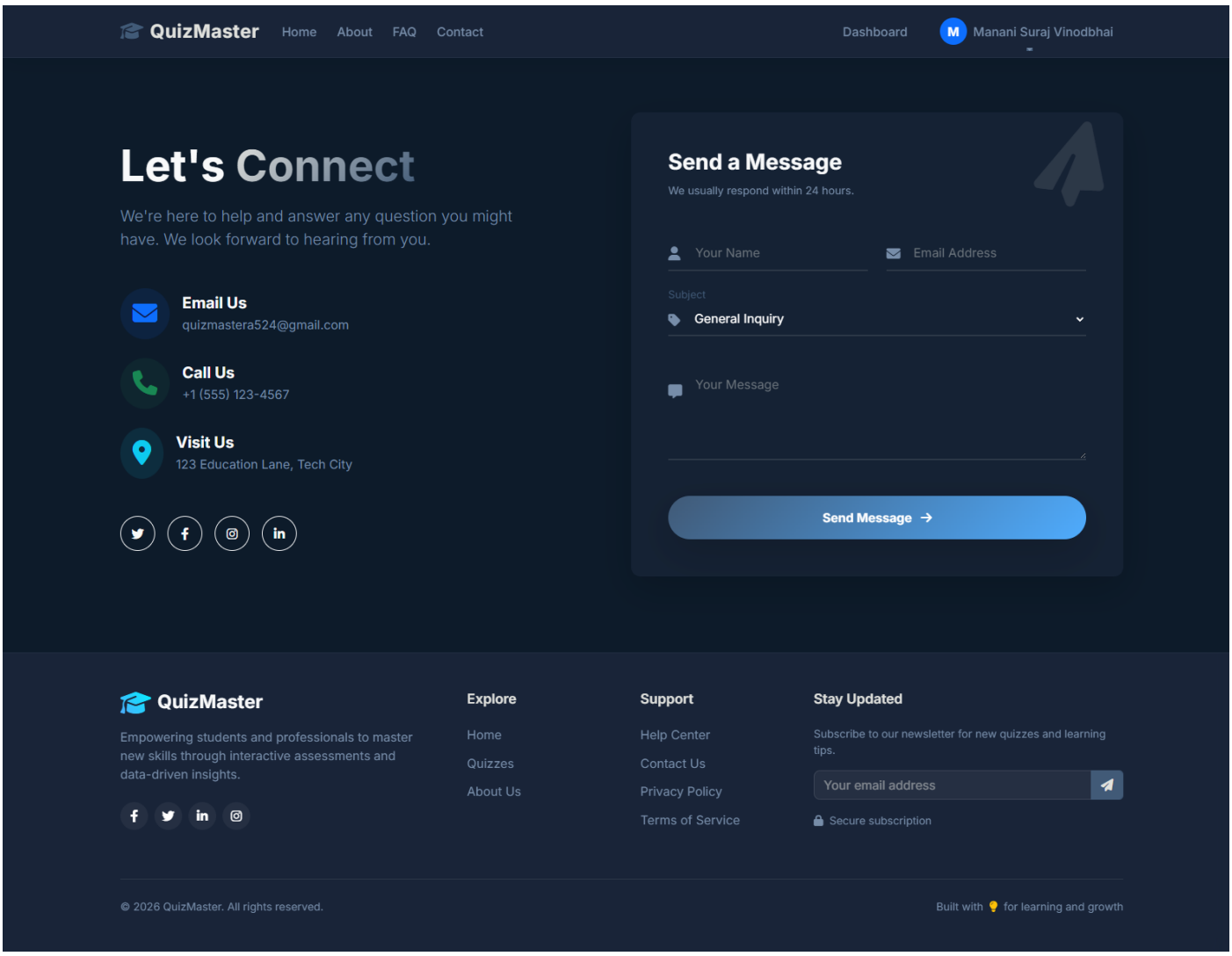
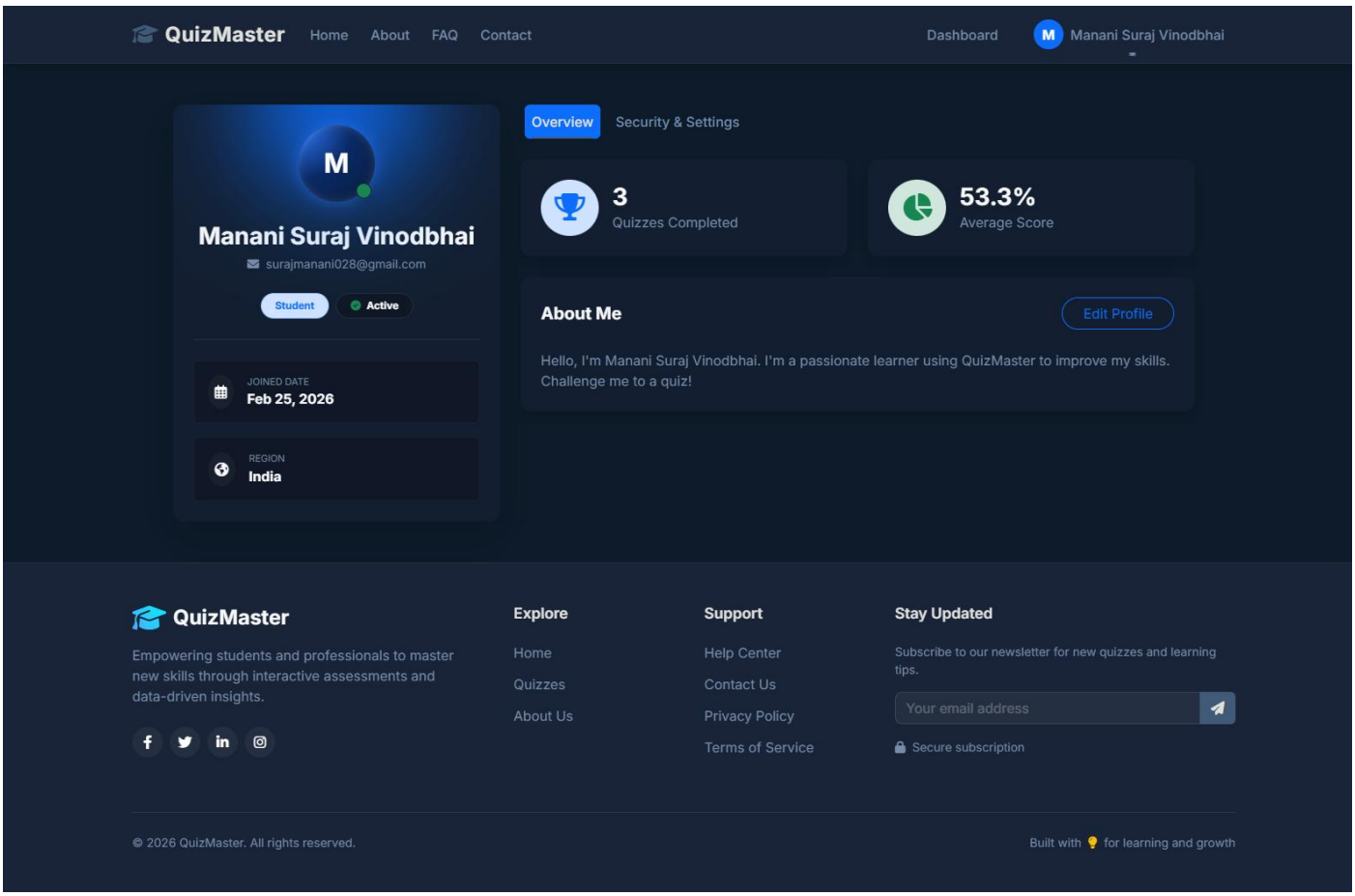
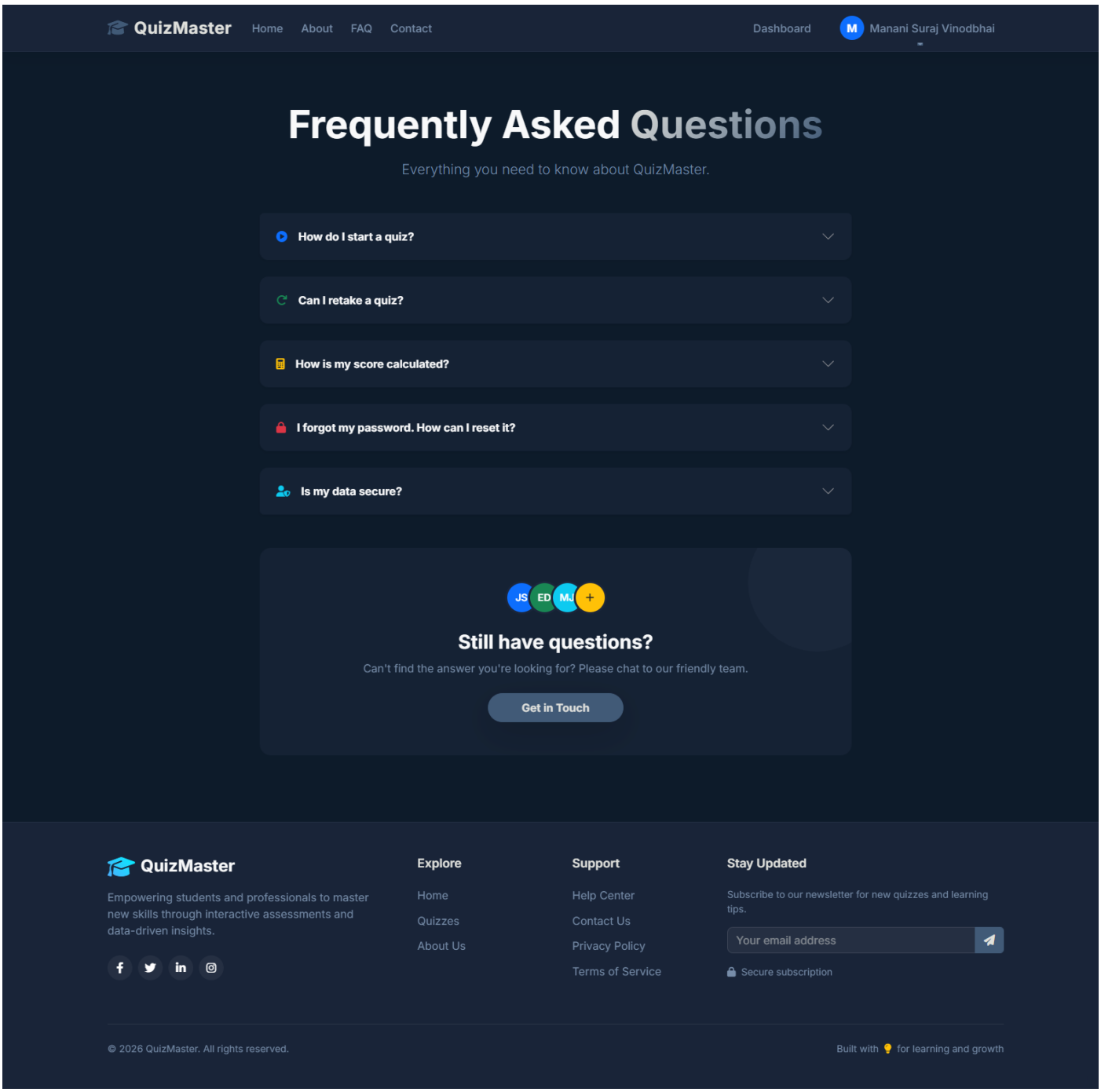


Profile Page:



FAQ Page: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Frequently Asked Questions
Everything you need to know about QuizMaster.
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Can I retake a quiz?
How is my score calculated?
I forgot my password. How can I reset it?
Is my data secure?
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Privacy Policy

How we handle and protect your data.

Last updated: March 02, 2026

1. Information We Collect

We collect information you provide directly to us when you create an account, participate in quizzes, or communicate with us. This may include your name, email address, password, and profile information. We also automatically collect certain information when you access our platform, such as your IP address, browser type, and usage data.

2. How We Use Your Information

We use the information we collect to operate, maintain, and improve our services. Specifically, we use your data to:

- Provide and personalize your quiz experience.
- Track your progress and generate performance reports.
- Send administrative notifications and updates.
- Detect and prevent fraudulent or unauthorized activity.

3. Data Security

We implement appropriate technical and organizational measures to protect specific data. However, please be aware that no method of transmission over the internet is completely secure. We strive to use commercially acceptable means to protect your personal information, but we cannot guarantee its absolute security.

4. Cookies

We use cookies and similar tracking technologies to track the activity on our service and hold certain information. Cookies are files with small amount of data which may include an anonymous unique identifier. You can instruct your browser to refuse all cookies or to indicate when a cookie is being sent.

5. Third-Party Services

We may employ third-party companies and individuals to facilitate our service, to provide the service on our behalf, to perform service-related services or to assist us in analyzing how our service is used. These third parties have access to your Personal Data only to perform these tasks on our behalf and are obligated not to disclose or use it for any other purpose.

6. Contact Us

If you have any questions about this Privacy Policy, please contact us at support@quizmaster.com.

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Terms of Service

Rules and agreements for using QuizMaster.

Effective Date: March 02, 2026

1. Acceptance of Terms

By accessing or using QuizMaster, you agree to be bound by these Terms of Service and all applicable laws and regulations. If you do not agree with any of these terms, you are prohibited from using or accessing this site.

2. User Accounts

When you create an account with us, you must provide information that is accurate, complete, and current at all times. Failure to do so constitutes a breach of the Terms, which may result in immediate termination of your account on our service. You are responsible for safeguarding the password that you use to access the service and for any activities or actions under your password.

3. Educational Content

The quizzes and educational materials provided on QuizMaster are for learning and assessment purposes. While we strive for accuracy, we do not guarantee that all content is free of errors. Report any discrepancies to our support team for review.

4. Prohibited Conduct

You agree not to use the service to:

- Violate any local, state, national, or international law.
- Attempt to exploit vulnerabilities or bypass authentication measures.
- Cheating or manipulating quiz scores through automated means.
- Harass, abuse, or harm another person or group.

5. Intellectual Property

The Service and its original content, features, and functionality are and will remain the exclusive property of QuizMaster and its licensors. The Service is protected by copyright, trademark, and other laws.

6. Termination

We may terminate or suspend access to our Service immediately, without prior notice or liability, for any reason whatsoever, including without limitation if you breach the Terms. All provisions of the Terms which by their nature should survive termination shall survive termination.

7. Changes to Terms

We reserve the right, at our sole discretion, to modify or replace these Terms at any time. What constitutes a material change will be determined at our sole discretion. By continuing to access or use our Service after those revisions become effective, you agree to be bound by the revised terms.

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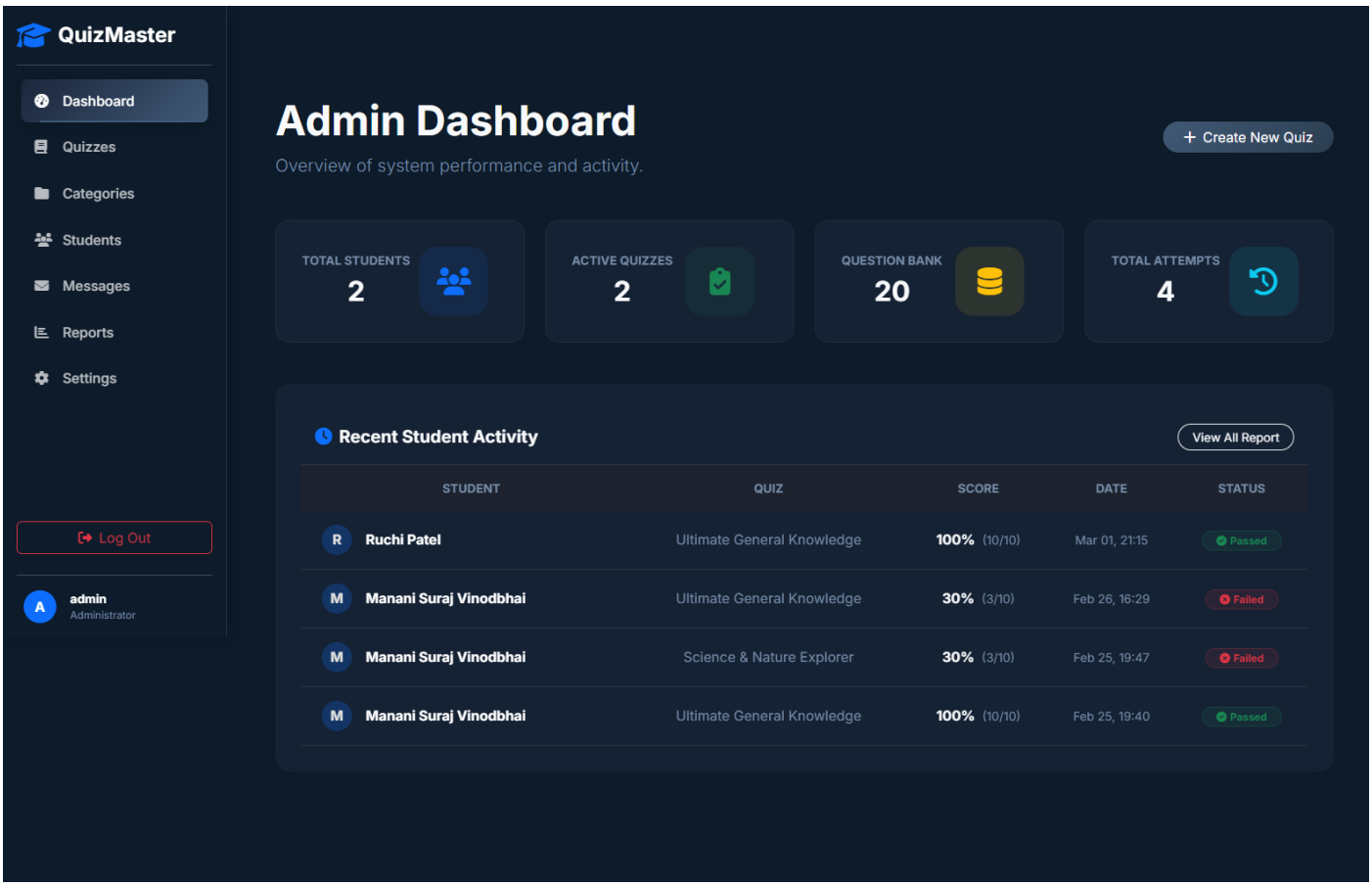
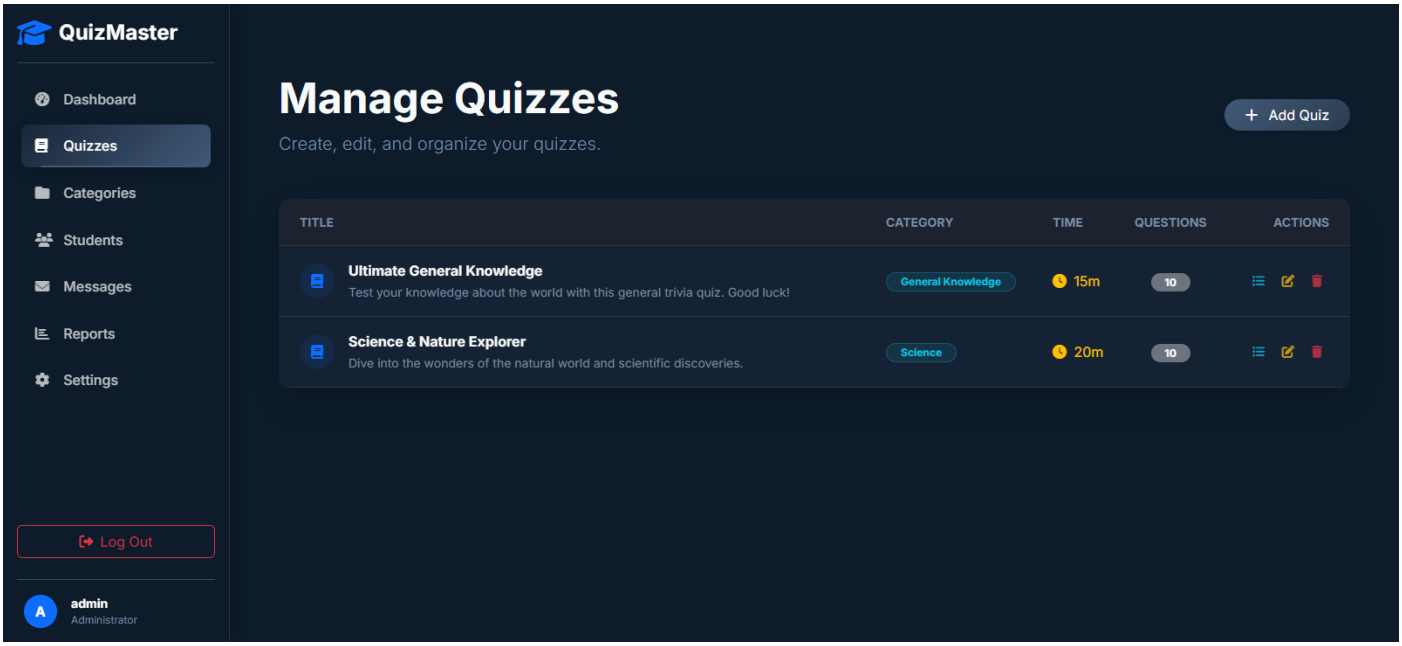
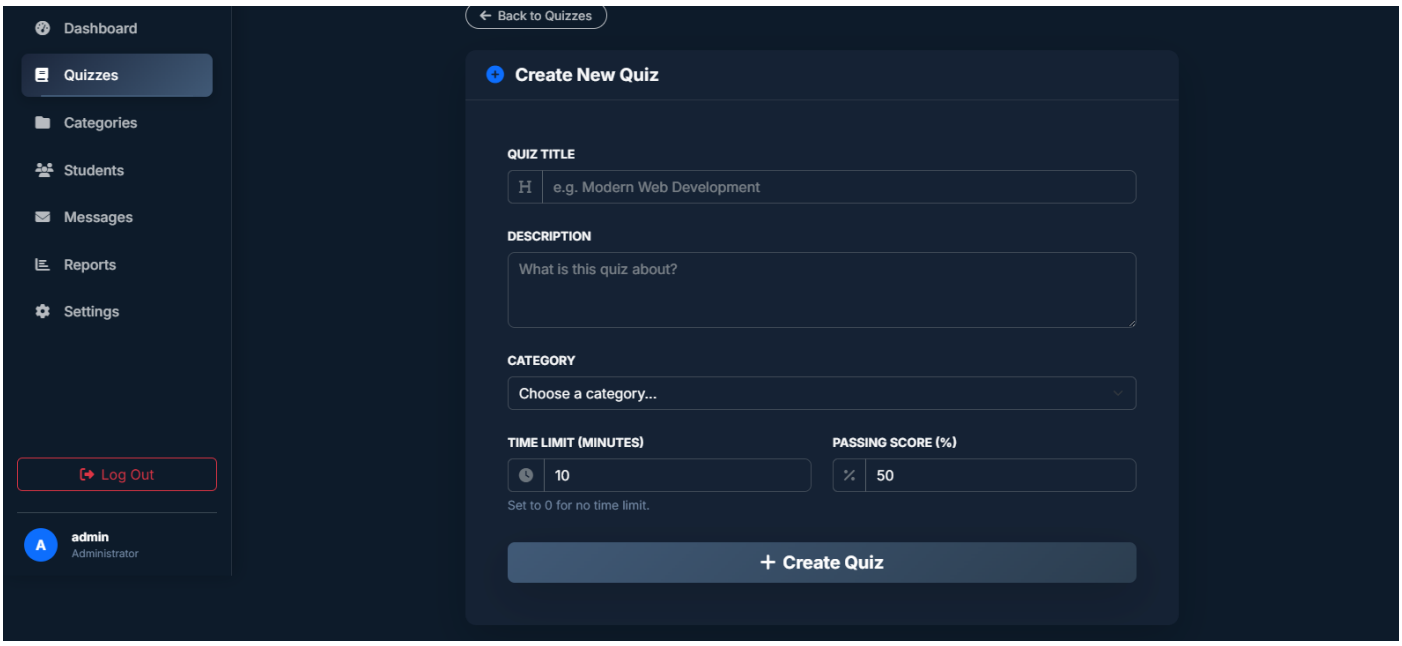
Admin Dashboard:



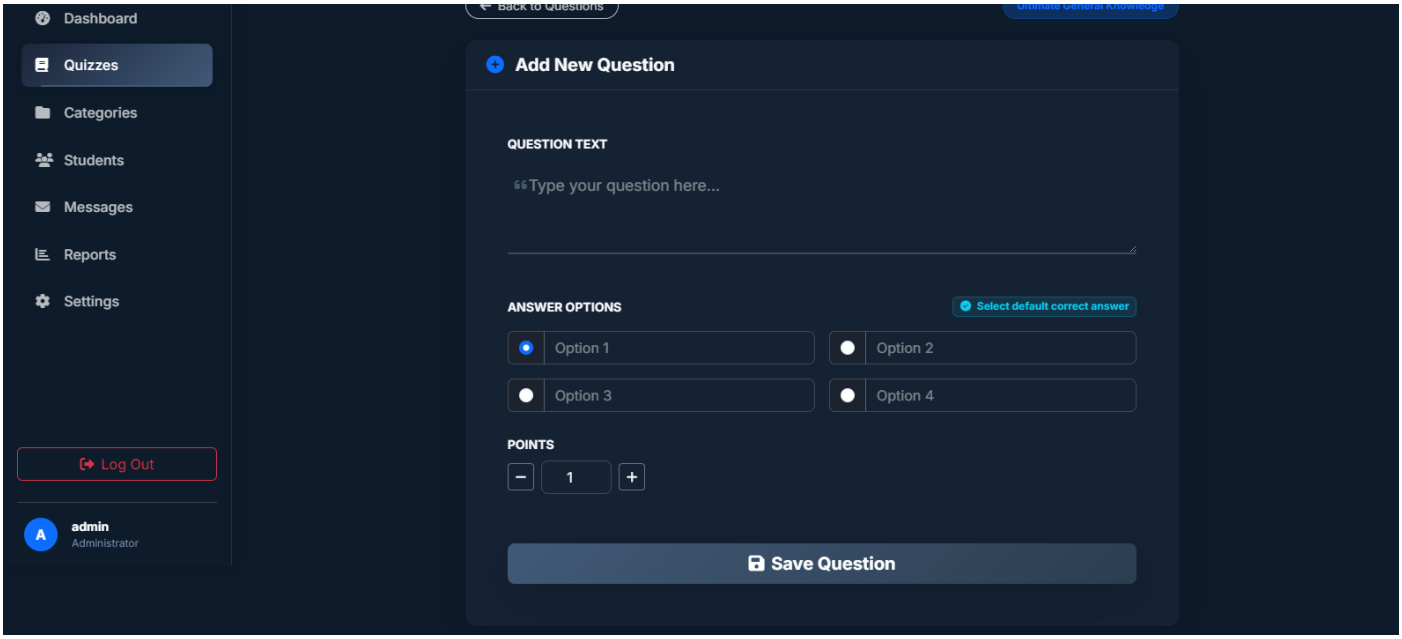
Manage Quiz:



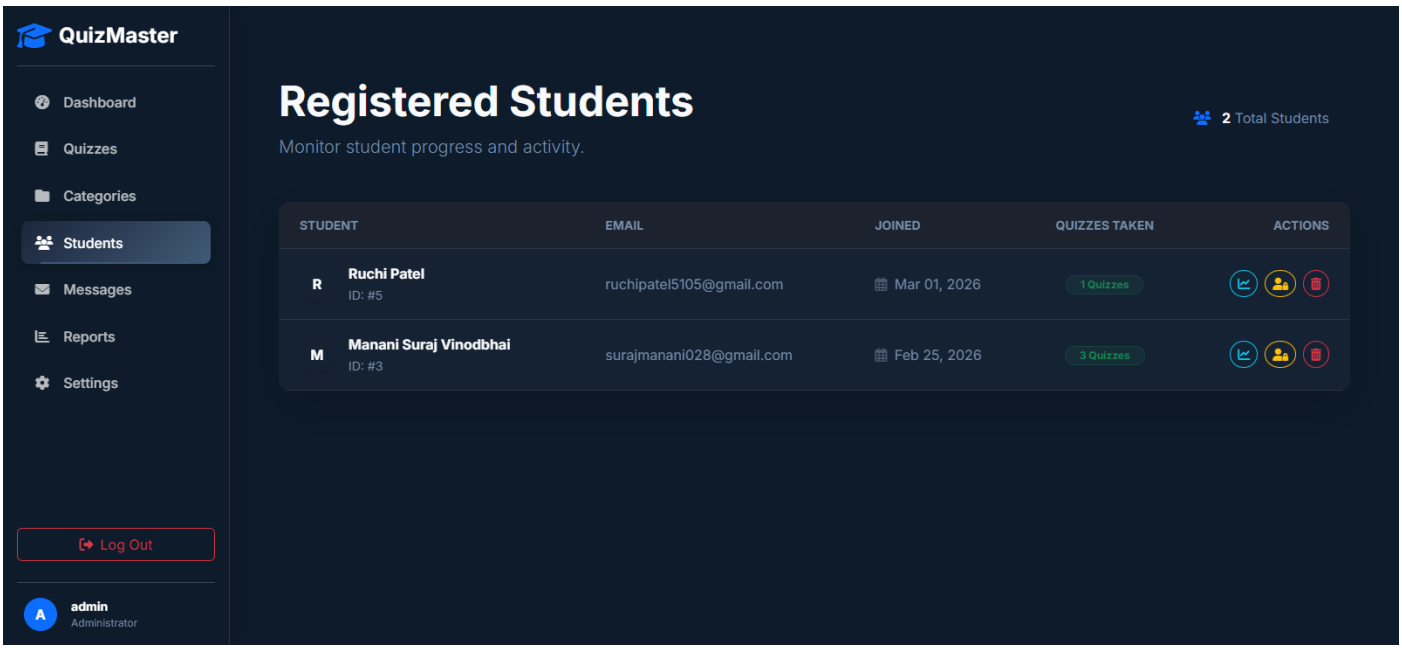
Add Quiz:



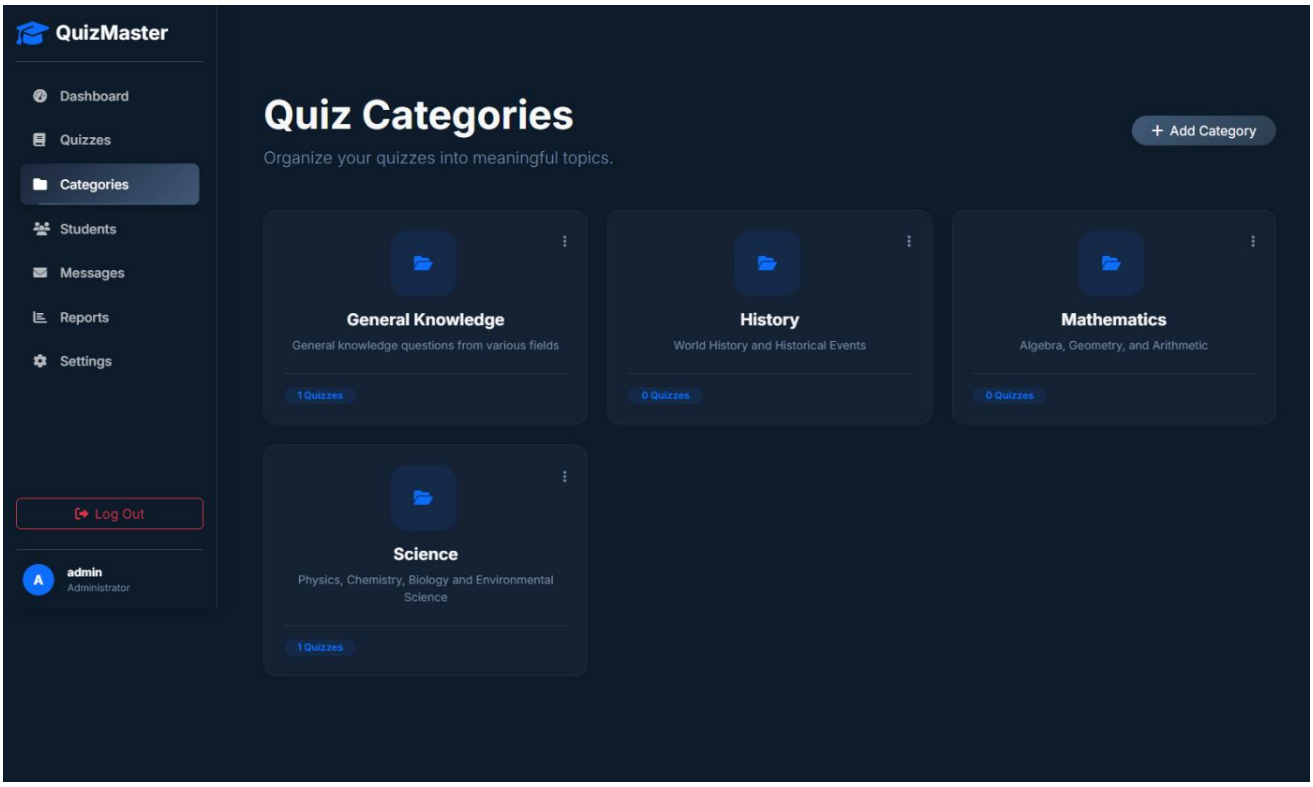
Add Questions:



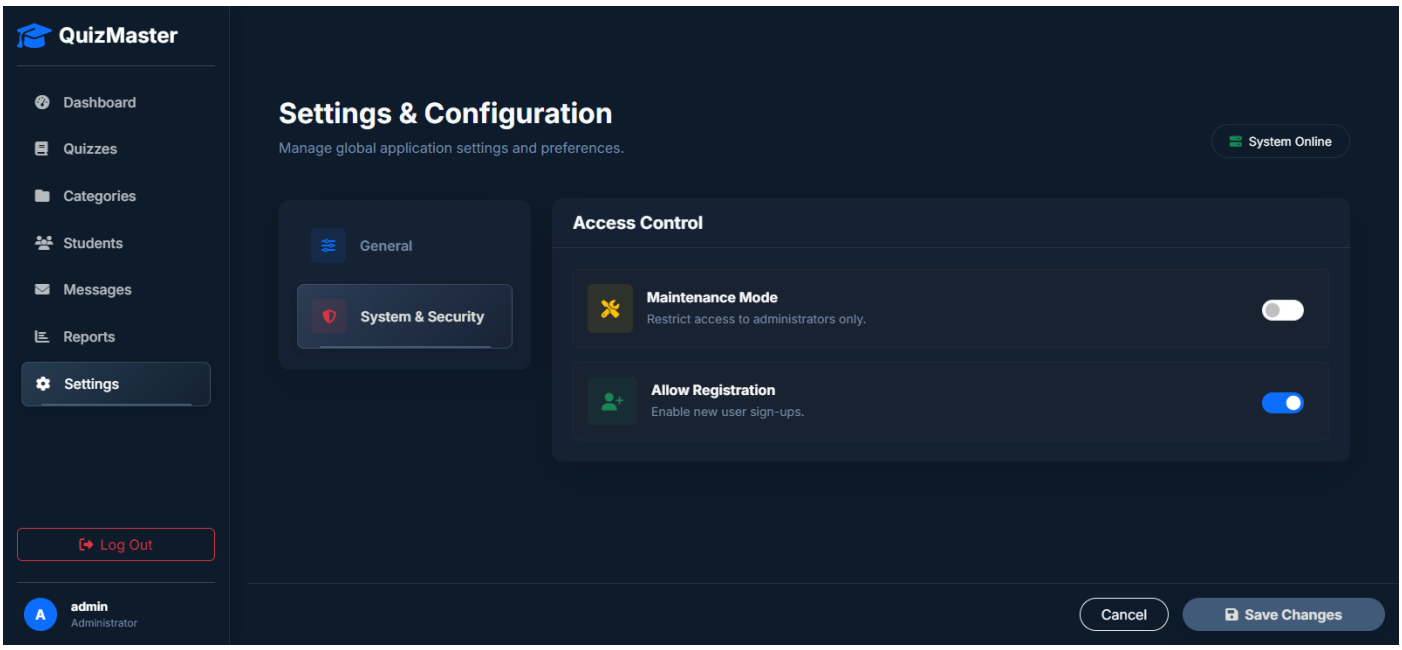
Manage Users:



Manage Category:



Setting Page:



Chapter 4:
System Testing

4.1 Fronted (Client-Side) Validation
4.2 Backend (Server-Side) Validation
4.3 Authentication & Authorization Validation
4.4 Manual Testing

Validations:

To ensure the application is secure and reliable, "QuizMaster" uses a multi-layered validation strategy. This serves as the primary method of system testing and is broken down into three layers.

4.1 Frontend (Client-Side) Validation:

Frontend validation provides immediate visual feedback to the user, ensuring a smooth experience before data even hits the PHP server. In QuizMaster, this is implemented using Vanilla JavaScript integrated deeply with Bootstrap 5's validation classes (`was-validated`).

When a user attempts to submit a form (e.g., Registration), the JavaScript intercepts the event. It checks for **Required Fields**, ensuring no empty data is sent. It also enforces format rules; for instance, **Email Validation** uses regex to ensure the input resembles `user@domain.com`. Furthermore, the registration script includes **Cross-Field Validation** to ensure the "Password" and "Confirm Password" fields are identical before allowing the form to proceed. This fail-fast approach prevents frustrating server-reload errors for simple typos.

4.2 Backend (Server-Side) Validation:

Backend Validation is the critical security gatekeeper. Because frontend JS can be bypassed, PHP must rigorously re-validate every incoming `$_POST` and `$_GET` request.

QuizMaster utilizes a central `sanitize()` function (found in `includes/functions.php`) to strip HTML tags and prevent XSS attacks across all text inputs. Furthermore, the database interactions rely exclusively on **PDO Prepared Statements**. Whether an admin is defining a quiz title, or a student is submitting an answer option, the data is bound as a parameter rather than concatenated into the SQL string. This completely neutralizes SQL Injection threats.

Additionally, the backend enforces **Logical Validation**. For example, the `take-quiz-process.php` script verifies that a submitted `option_id` actually belongs to the correct `question_id` in the database, preventing malicious users from tampering with hidden form fields to score unearned points.

4.3 Authentication & Authorization Validation:

Security in QuizMaster is managed through session-based authentication. When a user logs in, their password is securely validated against the bcrypt-hashed string in the database using PHP's `password_verify()` function. Upon success, a secure `$_SESSION['user_id']` and `$_SESSION['role']` are established.

Authorization is enforced via strict, included middleware functions like `requireLogin()` and `requireAdmin()`. Placed at the very top of sensitive scripts, these functions check the session state. If a student attempts to load `admin/dashboard.php`, the `requireAdmin()` function immediately halts execution and redirects them, ensuring Role-Based Access Control (RBAC) is tightly enforced.

The secure **Password Reset flow** utilizes unique, cryptographically secure random bytes (`bin2hex(random_bytes(32))`) to generate tokens. These tokens are saved to the database alongside a strict 15-minute MySQL timestamp expiration (`DATE_ADD(NOW(), INTERVAL 15 MINUTE)`). This ensures recovery links delivered via email are single-use and highly secure against interception.

4.4 Manual Testing:

Manual testing verifies the end-to-end functionality. Below is a summary of the executed test cases confirming the core features are stable and reliable for production use.

Test Case ID	Module	Test Description	Expected Result	Actual Result	Status
TC-01	Registration Module	User Registration	User record inserted into database and Welcome Email sent via PHPMailer	As Expected	Pass
TC-02	Authentication Module	User Login (Student)	Session created successfully and redirected to Student Dashboard	As Expected	Pass
TC-03	Quiz Browsing Module	Search & Filter Quizzes	SQL query correctly filters quizzes using GET parameters and pagination works properly	As Expected	Pass
TC-04	Quiz Attempt Module	Complete Quiz (Pass Condition)	Score calculated correctly, database updated, confetti animation displayed, PDF Certificate available	As Expected	Pass
TC-05	Authorization Module	Admin Unauthorized Access	Non-admin users redirected from /admin/ routes	As Expected	Pass
TC-06	Password Recovery Module	Forgot Password Flow	Reset token generated, email sent, user can reset password within 15 minutes	As Expected	Pass
TC-07	Certificate Module	Generate PDF Certificate	Valid FPDF certificate generated with correct user data and dark theme layout	As Expected	Pass

Chapter 5:
Future Enhancement

5.1 Enhanced User Interactions & Engagement
5.2 Advanced Context Management
5.3 Improved User Experience & Personalization

5.1 Enhanced User Interaction and Engagement

Future iterations of QuizMaster will focus on adding more engaging testing formats. A primary feature planned is **Fill-in-the-Blanks / Short Answer questions**. This requires complex backend string-matching algorithms (potentially utilizing Levenshtein distance for minor spelling errors) to grade text rather than just distinct multiple-choice IDs.

We also plan to implement a **Live Leaderboard System**. By gamifying the experience further, tracking total XP, badges earned, and top scores across the entire platform, we can create a competitive environment that encourages repeated engagement and learning.

5.2 Advanced Content Management

Advanced Context Management involves utilizing data to improve the educational value of the platform. We plan to integrate **Detailed Assessment Analytics**. For the educator, the dashboard will visualize not just overall scores, but question-by-question breakdown charts. This data-driven insight will allow teachers to identify "problem questions" where a majority of students failed, indicating a topic that needs to be retaught.

5.3 Improved User Experience and Personalization

Improving the User Experience (UX) is an ongoing process. A critical next step is the addition of an active, **On-Screen Real-Time Timer** during quizzes. While the backend currently enforces time limits, adding a visual countdown clock via Javascript (syncing periodically with the server) that auto-submits the form when time runs out will add a necessary sense of urgency and realism to examination conditions.

Additionally, we aim to allow **Profile Customization**. Allowing users to upload custom profile avatars (handling secure file uploads and MIME type validation in PHP) will break away from default UI components and allow users greater expression on the platform.

Chapter 6: Bibliography

6.1 Limitation
6.2 Future Scope
6.3 Conclusion

6.1 Limitation

Despite the robustness of the current QuizMaster system, there are technical limitations in this native architecture. The most significant is the lack of a **Front-Controller / MVC Framework**. Because it relies on direct page scripts (e.g., `results.php`), scaling the application to hundreds of distinct views will result in repeated code and harder routing management compared to using a framework like Laravel.

Session Scalability is another limitation. Native PHP sessions write to temporary files on the server's hard drive. If deployed to a load-balanced, multi-server environment (AWS, etc.), sessions will fail unless reconfigured to store data in a centralized memory cache like Redis or Memcached.

Finally, while the **Email Integration** using PHPMailer is functional and excellent for MVP environments, it is currently "synchronous". When a user completes a quiz, they must wait for the SMTP server to send the email before the results page loads. In a heavy production environment, this should be offloaded to an asynchronous background worker queue (e.g., Beanstalkd, RabbitMQ) to guarantee instantaneous page loads.

6.2 Conclusion

QuizMaster successfully demonstrates the power and flexibility of a modern, data-driven PHP assessment architecture. By heavily upgrading the standard LAMP stack with real-world features like SMTP Email integrations, secure token cryptography, and on-the-fly PDF Generation (FPDF), the project delivers an exceptional, interactive user experience that meets the high standards of modern EdTech.

The project highlights the effectiveness of robust server-side processing combined with modern frontend aesthetics. The seamless integration of Bootstrap 5 and custom CSS allows for a bespoke Glassmorphism design that breaks the mold of traditional, boring testing software. The user interface is an active participant in the educational process, guiding users intuitively and rewarding them visually.

From a learning perspective, this project serves as a masterclass in secure Full-Stack Development. It tackles critical, real-world development challenges: defending against SQL injection via PDO, handling accurate timezone mathematics for expiring tokens, migrating client-side search logic to dynamic server-side SQL pagination, and marrying complex libraries (PHPMailer, FPDF) into native scripts without relying on heavier frameworks like Composer auto-loading.

In conclusion, QuizMaster is an advanced, deployment-ready prototype. It successfully meets its primary objectives of modernizing assessment, streamlining operations, and providing a secure, engaging environment. It stands as a testament to the enduring capabilities of PHP when applied thoughtfully and offers a robust, highly-impressive portfolio piece for modern web development.